

METHOD OF DIFFERENTIATION

1.0 INTRODUCTION :

The essence of calculus is the derivative. The derivative is the instantaneous rate of change of a function with respect to one of its variables. This is equivalent to finding the slope of the tangent line to the function at a point. Let us use the view of derivatives as tangents to motivate a geometric definition of the derivative.

1.1 GEOMETRICAL MEANING OF A DERIVATIVE :

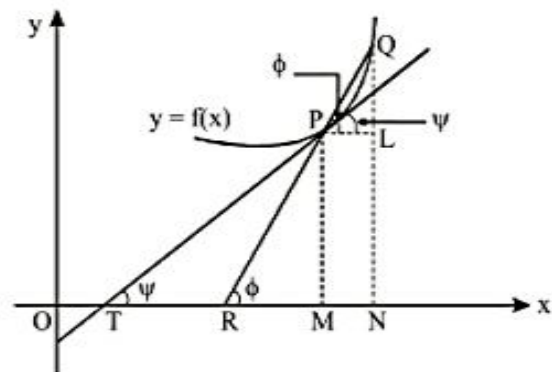
Let $P(x_0, f(x_0))$ and $Q(x_0 + h, f(x_0 + h))$ be two points very close to each other on the curve $y = f(x)$. Draw PM and QN perpendiculars from P and Q on x -axis, and draw PL as perpendicular from P on QN . Let the chord PQ produced meet the x -axis at R and $\angle QRN = \angle QPL = \phi$.

Now in right-angled triangle QPL

$$\begin{aligned}\tan \phi &= \frac{QL}{PL} = \frac{NQ - NL}{MN} = \frac{NQ - MP}{ON - OM} = \frac{f(x_0 + h) - f(x_0)}{(x_0 + h) - x_0} \\ &= \frac{f(x_0 + h) - f(x_0)}{h} \quad \dots\dots\dots(1)\end{aligned}$$

when $h \rightarrow 0$, the point Q moving along the curve tends to P , i.e., $Q \rightarrow P$. The chord PQ approaches the tangent line PT at the point P and then $\phi \rightarrow \psi$. Now applying $\lim_{h \rightarrow 0}$ in equation (1), we get

$$\begin{aligned}\lim_{h \rightarrow 0} \tan \phi &= \frac{f(x_0 + h) - f(x_0)}{h} \\ \tan \psi &= \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \\ \Rightarrow f'(x_0) &= \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}\end{aligned}$$



This definition of derivative is also called the first principle of derivative. Clearly, the domain of definition of $f'(x)$ is wherever the above limit exists.

Note that if $y = f(x)$ then the symbols

$$\frac{dy}{dx} = Dy = f'(x) = y_1 \text{ or } y' \text{ have the same meaning.}$$

However a dot, denotes the time derivative.

$$\text{e.g. } \dot{S} = \frac{dS}{dt} ; \quad \dot{\theta} = \frac{d\theta}{dt} \text{ etc.}$$

Illustration :

Find the derivative of $e^{\sqrt{x}}$ w.r.t. x using first principle.

Sol. Let $f(x) = e^{\sqrt{x}}$, then $f(x+h) = e^{\sqrt{x+h}}$

$$\begin{aligned}\therefore \frac{d}{dx} f(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{e^{\sqrt{x+h}} - e^{\sqrt{x}}}{h} = e^{\sqrt{x}} \lim_{h \rightarrow 0} \left(\frac{e^{\sqrt{x+h} - \sqrt{x}} - 1}{h} \right) \\ &= e^{\sqrt{x}} \lim_{h \rightarrow 0} \left(\frac{e^{\sqrt{x+h} - \sqrt{x}} - 1}{\sqrt{x+h} - \sqrt{x}} \right) \left(\frac{\sqrt{x+h} - \sqrt{x}}{h} \right) \\ &= e^{\sqrt{x}} \lim_{h \rightarrow 0} \left(\frac{e^{\sqrt{x+h} - \sqrt{x}} - 1}{\sqrt{x+h} - \sqrt{x}} \right) \times \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} \\ &= e^{\sqrt{x}} \lim_{h \rightarrow 0} \left(\frac{e^y - 1}{y} \right) \lim_{h \rightarrow 0} \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})}, \\ \text{where } y &= \sqrt{x+h} - \sqrt{x} \quad (\because \text{when } h \rightarrow 0, y \rightarrow 0) \\ &= e^{\sqrt{x}} \times 1 \times \left(\frac{1}{\sqrt{x} + \sqrt{x}} \right) = \frac{e^{\sqrt{x}}}{2\sqrt{x}}.\end{aligned}$$

Illustration :

Find the derivative of the following functions with respect to x using first principle.

(i) $y = \frac{x}{x^2+1}$; (ii) $y = \cos^2 x$; (iii) $y = \sin 3x$; (iv) $y = x^3 - 3^x$; (v) $y = \ln^2 x$.

Sol.

(i) $y = \frac{x}{x^2+1} = f(x)$

$$\begin{aligned}\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{x+h}{(x+h)^2+1} - \frac{x}{x^2+1}}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)(x^2+1) - x[(x^2+1) + 2hx + h^2]}{h(x^2+1)[(x+h)^2+1]} = \lim_{h \rightarrow 0} \frac{(x^2+1) - x(2x+h)}{(x^2+1)[(x+h)^2+1]} \\ &= \lim_{h \rightarrow 0} \frac{(x^2+1) - 2x^2}{(x^2+1)^2} = \lim_{h \rightarrow 0} \frac{1-x^2}{(x^2+1)^2}.\end{aligned}$$

(ii) $y = \cos^2 x = f(x)$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\cos^2(x+h) - \cos^2 x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 \sin\left(x + \frac{h}{2}\right) \sin\left(\frac{-h}{2}\right)}{h} (\cos(x+h) + \cos x)$$

$$= 2 \sin x \left(\frac{-1}{2}\right) (2 \cos x) = -2 \sin x \cos x = -\sin 2x.$$

(iii) $y = \sin 3x = f(x)$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(3x+3h) - \sin 3x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 \sin\left(\frac{3h}{2}\right) \cos\left(3x + \frac{3h}{2}\right)}{h} = 2 \cdot \left(\frac{3}{2}\right) \cdot \cos(3x+0) = 3 \cos 3x.$$

(iv) $y = x^3 - 3^x = f(x)$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^3 - 3^{x+h} - (x^3 - 3^x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} - \lim_{h \rightarrow 0} \frac{(3^{x+h} - 3^x)}{h} = \lim_{h \rightarrow 0} \frac{3xh(x+h) + h^3}{h} - 3^x \lim_{h \rightarrow 0} \frac{(3^h - 1)}{h}$$

$$= 3x(x+0) - 3^x \ln 3 = 3x^2 - 3^x \ln 3.$$

(v) $y = f(x) = \ln^2 x$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\ln^2(x+h) - \ln^2 x}{h}$$

$$= \lim_{h \rightarrow 0} \left(\frac{\ln(x+h) - \ln x}{h} \right) (\ln(x+h) + \ln x)$$

$$= \lim_{h \rightarrow 0} \frac{\ln\left(1 + \frac{h}{x}\right)}{h} (\ln(x+h) + \ln x) = \frac{2 \ln x}{x}.$$

1.2 STANDARD DERIVATIVES :

$$(i) \quad \frac{d}{dx} x^n = nx^{n-1}, x \in \mathbb{R}, n \in \mathbb{R}, x > 0$$

$$(ii) \quad \frac{d}{dx} (e^x) = e^x$$

$$(iii) \quad \frac{d}{dx} (a^x) = a^x \ln a$$

$$(iv) \quad \frac{d}{dx} (\ln |x|) = \frac{1}{x}$$

$$(v) \quad \frac{d}{dx} (\log_a x) = \frac{1}{x} \log_a e$$

$$(vi) \quad \frac{d}{dx} (\sin x) = \cos x$$

$$(vii) \quad \frac{d}{dx} (\cos x) = -\sin x$$

$$(viii) \quad \frac{d}{dx} (\tan x) = \sec^2 x$$

$$(ix) \quad \frac{d}{dx} (\cot x) = -\operatorname{cosec}^2 x$$

$$(x) \quad \frac{d}{dx} (\sec x) = \sec x \tan x$$

$$(xi) \quad \frac{d}{dx} (\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

Illustration :

If $y = \left(1 + x^{\frac{1}{4}}\right) \left(1 + x^{\frac{1}{2}}\right) \left(1 - x^{\frac{1}{4}}\right)$, then find $\frac{dy}{dx}$.

$$\begin{aligned} \text{Sol. } y &= \left(1 + x^{\frac{1}{4}}\right) \left(1 + x^{\frac{1}{2}}\right) \left(1 - x^{\frac{1}{4}}\right) = \left(1 + x^{\frac{1}{4}}\right) \left(1 - x^{\frac{1}{4}}\right) \left(1 + x^{\frac{1}{2}}\right) = \left(1 - x^{\frac{1}{2}}\right) \left(1 + x^{\frac{1}{2}}\right) \\ &= 1 - x \Rightarrow \frac{dy}{dx} = -1. \end{aligned}$$

Illustration :

If $f(x) = x |x|$, then prove that $f'(x) = 2|x|$.

$$\begin{aligned} \text{Sol. } f(x) &= \begin{cases} -x^2, & x < 0 \\ x^2, & x > 0 \end{cases} \Rightarrow f'(x) = \begin{cases} -2x, & x < 0 \\ 2x, & x > 0 \end{cases} \\ \therefore f'(x) &= 2|x|. \end{aligned}$$

Illustration :

If $y = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}$, $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$, then find $\frac{dy}{dx}$.

Sol. We have

$$\begin{aligned} y &= \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}} = \sqrt{\frac{2 \sin^2 x}{2 \cos^2 x}} = \sqrt{\tan^2 x} \Rightarrow y = |\tan x|, \text{ where } x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right) \\ \Rightarrow y &= \begin{cases} \tan x, & x \in \left(0, \frac{\pi}{2}\right) \\ -\tan x, & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases} \Rightarrow \frac{dy}{dx} = \begin{cases} \sec^2 x, & \text{if } x \in \left(0, \frac{\pi}{2}\right) \\ -\sec^2 x, & \text{if } x \in \left(\frac{\pi}{2}, \pi\right) \end{cases} \end{aligned}$$

Illustration :

Find the derivative of the following functions

- (i) $x \sin x$; (ii) $e^x \cdot \tan x$

Sol.

(i) $y = x \sin x$

$$y = x \sin x \text{ so } \frac{dy}{dx} = \sin x \frac{d}{dx}(x) + x \frac{d}{dx}(\sin x)$$

$$\frac{dy}{dx} = \sin x + x \cos x$$

(ii) $y = e^x \tan x$

$$\text{so } \frac{dy}{dx} = \tan x \frac{d}{dx}(e^x) + e^x \frac{d}{dx}(\tan x) = e^x (\tan x + \sec^2 x)$$

Note : If 3 functions are involved then remember

$$\begin{aligned} D(f(x) \cdot g(x) \cdot h(x)) &= f(x) \cdot g(x) \cdot h'(x) + g(x) \cdot h(x) \cdot f'(x) + h(x) \cdot f(x) \cdot g'(x) \\ &= \frac{(fg)'(h) + (gh)'(f) + (hf)'(g)}{2} \end{aligned}$$

This result can be generalised to the product of n functions

Illustration :

Let $F(x) = f(x) \cdot g(x) \cdot h(x)$. If for some $x = x_0$, $F'(x_0) = 21$, $F(x_0) = 21$, $f'(x_0) = 4f(x_0)$; $g'(x_0) = -7g(x_0)$ and $h'(x_0) = k h(x_0)$ then find k.

Sol. $F(x) = f(x) g(x) h(x)$

Given that $F'(x_0) = 21f(x_0)$, $f'(x_0) = 4f(x_0)$; $g'(x_0) = -7g(x_0)$ and $h'(x_0) = kh(x_0)$

$$F'(x_0) = f'(x_0) g(x_0) h(x_0) + f(x_0) g'(x_0) h(x_0) + f(x_0) g(x_0) h'(x_0)$$

$$21F(x_0) = (4 - 7 + k) f(x_0) g(x_0) h(x_0) \Rightarrow 21 = -3 + k \Rightarrow k = 24.$$

Illustration :

Let f , g and h are differentiable functions. If $f(0) = 1$; $g(0) = 2$; $h(0) = 3$ and the derivatives of their pair wise products at $x = 0$ are

$$(fg)'(0) = 6; (gh)'(0) = 4 \text{ and } (hf)'(0) = 5$$

then compute the value of $(fgh)'(0)$.

Sol. Let $w(x) = f(x) g(x) h(x)$

$$\frac{d}{dx}(w(x)) = f'(x) g(x) h(x) + f(x) g'(x) h(x) + f(x) g(x) h'(x)$$

$$2 \frac{d}{dx}(w(x)) = 2f'(x) g(x) h(x) + 2f(x) g'(x) h(x) + 2f(x) g(x) h'(x)$$

$$= h(x) \frac{d}{dx}(f(x) g(x)) + g(x) \frac{d}{dx}(f(x) h(x)) + f(x) \frac{d}{dx}(g(x) h(x))$$

$$2 \frac{d}{dx}(w(0)) = (3)(6) + (2)(5) + (1)(4) = 32$$

$$\frac{d}{dx}(w(0)) = 16.$$

Illustration :

If $f(x) = 1 + x + x^2 + \dots + x^{100}$ then $f'(1) = \underline{\hspace{2cm}}$.

Sol. $f(x) = 1 + x + x^2 + \dots + x^{100}$
 $f'(x) = 0 + 1 + 2x + 3x^2 + \dots + 100x^{99}$

$$\therefore f'(1) = 1 + 2 + 3 + \dots + 100 = \frac{100(100+1)}{2} = 5050$$

T-4 QUOTIENT RULE :

$$y = \frac{f(x)}{g(x)}$$

$$D\left(\frac{f(x)}{g(x)}\right) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{g^2(x)}, \text{ to be remembered as}$$

$$D\left(\frac{N^r}{D^r}\right) = \frac{D^r \frac{d}{dx}(N^r) - N^r \frac{d}{dx}(D^r)}{(D^r)^2};$$

Note: If $y = \frac{1}{f(x)}$ then $D(y) = -\frac{f'(x)}{f^2(x)}$

Illustration :

Find the derivative of the following functions

(i) $y = \frac{1 - \ln x}{1 + \ln x}$; (ii) $y = \frac{x^3 + 2^x}{e^x}$; (iii) $y = \frac{x \sin x}{1 + \tan x}$

Sol.

(i) $y = \frac{1 - \ln x}{1 + \ln x}$

$$\frac{dy}{dx} = \frac{(1 + \ln x) \left(\frac{-1}{x} \right) - (1 - \ln x) \cdot x}{(1 + \ln x)^2} = \frac{-(1 + \ln x + 1 - \ln x)}{x(1 + \ln x)^2} = \frac{-2}{x(1 + \ln x)^2}$$

(ii) $y = \frac{x^3 + 2^x}{e^x} = x^3 e^{-x} + 2^x e^{-x}$

$$\frac{dy}{dx} = 3x^2 e^{-x} - x^3 e^{-x} - 2^x e^{-x} + 2^x e^{-x} (\ln 2) = (3x^2 - x^3 - 2^x + 2^x \ln 2) e^{-x}$$

$$(iii) \quad y = \frac{x \sin x}{(1 + \tan x)}$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{(1 + \tan x)(\sin x + x \cos x) - (x \sin x) \sec^2 x}{(1 + \tan x)^2} \\ &= \frac{\sin x + x \cos x + \sin x \tan x + x \sin x - x \tan x \sec x}{(1 + \tan x)^2} \\ &= \frac{(1 + x) \sin x + x \cos x + (\sin x - x \sec x) \tan x}{(1 + \tan x)^2} \end{aligned}$$

Illustration :

$$(i) \quad \text{If } y = \frac{x^4 + x^2 + 1}{x^2 + x + 1} \text{ then } \frac{dy}{dx} = ax + b \text{ find } a \text{ and } b.$$

$$(ii) \quad \text{If } y = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1}, \text{ find } \left. \frac{dy}{dx} \right|_{x=\pi/4}$$

$$(iii) \quad \text{If } y = \frac{x^3 + x^2 + x}{1 + x^2}, \text{ find } \frac{dy}{dx}.$$

Sol.

$$(i) \quad y = \frac{x^4 + x^2 + 1}{x^2 + x + 1} = \frac{(x^4 + 2x^2 + 1) - x^2}{x^2 + x + 1} = x^2 - x + 1$$

$$\frac{dy}{dx} = 2x - 1 = ax + b \Rightarrow a = 2, b = -1$$

$$(ii) \quad y = \frac{\sec x + \tan x - 1}{\tan x - \sec x + 1} = \frac{\sec x + \tan x - (\sec^2 x - \tan^2 x)}{\tan x - \sec x + 1}$$

$$y = \frac{(\sec x + \tan x)(1 - \sec x + \tan x)}{(\tan x - \sec x + 1)}$$

$$y = \sec x + \tan x$$

$$\therefore \frac{dy}{dx} = \sec x \tan x + \sec^2 x = \sec x (\tan x + \sec x)$$

$$(iii) \quad y = \frac{x^3 + x^2 + x}{1 + x^2} = (x + 1) - \frac{1}{x^2 + 1}$$

$$\frac{dy}{dx} = 1 - \frac{(x^2 + 1) \frac{d}{dx}(1) - 1 \cdot \frac{dy}{dx}(x^2 + 1)}{(x^2 + 1)^2} = 1 - \frac{(0 - 2x)}{(x^2 + 1)^2} = 1 + \frac{2x}{(x^2 + 1)^2}.$$

T-5 DIFFERENTIATION OF COMPOSITE FUNCTION (CHAIN RULE) :

If $f(x)$ and $g(x)$ are differentiable functions, then $f \circ g$ is also differentiable and $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$.

$$\text{or, } \frac{d}{dx} \{f \circ g(x)\} = \frac{d}{d g(x)} \{(f \circ g)(x)\} \frac{d}{dx} (g(x))$$

or

If y is a function of t and t is a function of x , then

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

Thus, if $y = f(t)$ and $t = \phi(x)$

$$\frac{dt}{dx} = \phi'(x)$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = f'(t) \phi'(x). \text{ This rule is called Chain Rule.}$$

This chain rule can be extended as follows

Let $y = f(t)$, $t = \phi(z)$, $z = \psi(x)$

$$\text{then } \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dz} \times \frac{dz}{dx} = f'(t) \phi'(z) \psi'(x)$$

Let $y = \log \sin x^3 = \log t$

Putting $t = \sin x^3 = \sin z$, $z = x^3$

$$\Rightarrow \frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dz} \times \frac{dz}{dx} = (1/t) \cos z \cdot 3x^2 = \left(\frac{1}{\sin x^3} \right) (\cos x^3) \times 3x^2 = 3x^2 \cot x^3$$

Note : If $y = f(u)$ and $u = g(x)$ then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$.

It can be extended for any number of chain. In general if $y = (E)^C$ where E = function of x and C = constant then

$$\frac{dy}{dx} = C(E)^{C-1} \cdot \frac{d}{dx}(E)$$

Illustration :

Find derivative of following functions ?

- (i) $y = \sin^3 \sqrt{x}$ (ii) $y = \ln (\sec x)$ (iii) $y = \cos(\ln x)$
(iv) $y = e^{ax} \sin bx$ (v) $y = e^{ax} \cos bx$

Sol.

(i) $y = \sin^3 \sqrt{x}$

Let $v = \sqrt{x}$; $u = \sin \sqrt{x} = \sin v$
 $y = u^3$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(u^3) = \frac{d}{dx}(u^3) \cdot \left(\frac{du}{dx} \right) = 3u^2 \cdot \frac{d}{dv}(\sin v) \left(\frac{dv}{dx} \right) = 3u^2 \cos v \frac{d}{dx}(\sqrt{x}) \\ &= 3 \sin^2 v \cos v \left(\frac{1}{2\sqrt{x}} \right) = \frac{3 \sin^2 \sqrt{x} \cdot \cos \sqrt{x}}{2\sqrt{x}} \end{aligned}$$

(ii) $y = \ln (\sec x)$

Let $u = \sec x$

$\therefore y = \ln u$

$$\frac{dy}{dx} = \frac{d}{dx} (\ln u) = \frac{d}{du} (\ln u) \frac{du}{dx} = \frac{1}{u} \cdot \frac{d}{dx} (\sec x) = \frac{1}{\sec x} (\sec x \tan x) = \tan x$$

(iii) $y = \cos (\ln x)$

Let $u = \ln x$

$\therefore y = \cos u$

$$\frac{dy}{dx} = \frac{d}{dx} (\cos u) = \frac{d}{du} (\cos u) \frac{du}{dx} = -\sin u \cdot \frac{d}{dx} (\ln x) = \frac{-\sin(\ln x)}{x}$$

(iv) $y = e^{ax} \sin bx$

$$\begin{aligned} \frac{dy}{dx} &= e^{ax} \frac{d}{dx} (\sin bx) + \sin bx \frac{d}{dx} e^{ax} \\ &= e^{ax} \cos bx \frac{d}{dx} (bx) + \sin bx \cdot e^{ax} \frac{d}{dx} (ax) \\ &= be^{ax} \cos bx + a \sin bx \cdot e^{ax} \\ &= (b \cos bx + a \sin bx) e^{ax} \end{aligned}$$

(v) $y = e^{ax} \cos bx$

$$\begin{aligned} \frac{dy}{dx} &= e^{ax} \frac{d}{dx} (\cos bx) + \cos bx \frac{d}{dx} (e^{ax}) = e^{ax} (-\sin bx) \frac{d}{dx} (bx) + \cos bx e^{ax} \cdot a \\ &= (a \cos bx - b \sin bx) e^{ax} \end{aligned}$$

Illustration :

Find derivative of following functions ?

(i) $y = (f \circ g)(x)$	(ii) $y = (g \circ f)(x)$	(iii) $y = \frac{1}{(f(x))^n}$
(iv) $y = \sec^2(f^3(x))$	(v) $y = \sqrt{f(x)}$	(vi) $y = f(1/x)$

Sol.

(i) $y = f \circ g(x) = f(g(x))$

$$\frac{dy}{dx} = \frac{d}{dx} f(g(x)) = f'(g(x)) \cdot \frac{d}{dx} g(x) = f'(g(x)) \cdot g'(x)$$

(ii) $y = g \circ f(x) = g(f(x))$

$$\frac{dy}{dx} = \frac{d}{dx} g(f(x)) = g'(f(x)) \cdot \frac{d}{dx} f(x) = g'(f(x)) \cdot f'(x)$$

(iii) $y = \frac{1}{(f(x))^n} = (f(x))^{-n}$

$$\frac{dy}{dx} = -n (f(x))^{-n-1} \frac{d}{dx} f(x) = \frac{-n f'(x)}{(f(x))^{n+1}}$$

(iv) $y = \sec^2 (f^3(x))$

$$\begin{aligned}\frac{dy}{dx} &= 2 \sec (f^3(x)) \frac{d}{dx} \sec (f^3(x)) \\ &= 2 \sec (f^3(x)) \cdot \sec (f^3(x)) \tan (f^3(x)) \frac{d}{dx} (f^3(x)) \\ &= 2 \sec^2 (f^3(x)) \tan (f^3(x)) 3f^2(x) \frac{d}{dx} f(x) \\ &= 6 \sec^2 (f^3(x)) \tan (f^3(x)) f^2(x) \cdot f'(x)\end{aligned}$$

(v) $y = \sqrt{f(x)}$

$$\therefore \frac{dy}{dx} = \frac{1}{2} (f(x))^{-\frac{1}{2}} \frac{d}{dx} f(x) = \frac{f'(x)}{2\sqrt{f(x)}}$$

(vi) $y = f\left(\frac{1}{x}\right)$

$$\therefore \frac{dy}{dx} = f'\left(\frac{1}{x}\right) \frac{d}{dx} \left(\frac{1}{x}\right) = f'\left(\frac{1}{x}\right) \left(-\frac{1}{x^2}\right) = -\frac{f'\left(\frac{1}{x}\right)}{x^2}$$

Illustration :

Find derivative of following functions ?

(i) $y = \ln^3 \tan^2(x^4)$ (ii) $y = \cos\left(\frac{ax}{b}\right)$ (iii) $y = e^{\sqrt{\sin(\ln(x^2+7))}}$

(iv) $y = \sec x \sqrt{\tan x}$ (v) $\exp(\cos^3(\tan^{-1}x^3)^2)$

Sol.

(i) $y = \ln^3 (\tan^2(x^4))$

$$\begin{aligned}\frac{dy}{dx} &= 3 \ln^2 (\tan^2(x^4)) \frac{d}{dx} \ln (\tan^2(x^4)) \\ &= 3 \ln^2 (\tan^2(x^4)) \frac{1}{\tan^2(x^4)} \frac{d}{dx} \tan^2(x^4) \\ &= \frac{3 \ln^2 (\tan^2(x^4))}{\tan^2(x^4)} \cdot 2 \tan(x^4) \cdot \frac{d}{dx} \tan(x^4) \\ &= \frac{6 \ln^2 (\tan^2(x^4))}{\tan(x^4)} \sec^2(x^4) \frac{d}{dx} (x^4) \\ &= \frac{24x^3 \ln^2 (\tan^2(x^4)) \sec^2(x^4)}{\tan(x^4)}\end{aligned}$$

$$(ii) \quad y = \cos \left(\frac{ax}{b} \right)$$

$$\frac{dy}{dx} = -\sin \left(\frac{ax}{b} \right) \frac{d}{dx} \left(\frac{ax}{b} \right) = \frac{-a \sin \left(\frac{ax}{b} \right)}{b}$$

$$(iii) \quad y = e^{\sqrt{\sin(\ln(x^2+7))}}$$

$$\therefore y = e^{\sqrt{\sin(5 \ln(x^2+7))}}$$

$$\therefore \frac{dy}{dx} = e^{\sqrt{\sin(5 \ln(x^2+7))}} \cdot \frac{d}{dx} \sqrt{\sin(5 \ln(x^2+7))}$$

$$= e^{\sqrt{\sin(5 \ln(x^2+7))}} \cdot \frac{1}{2\sqrt{\sin(5 \ln(x^2+7))}} \cdot \frac{d}{dx} \sin(5 \ln(x^2+7))$$

$$= \frac{e^{\sqrt{\sin(5 \ln(x^2+7))}}}{2\sqrt{\sin(5 \ln(x^2+7))}} \cos(5 \ln(x^2+7)) \cdot \frac{5}{(x^2+7)} \cdot \frac{d}{dx} (x^2+7)$$

$$= \frac{5x \cos(5 \ln(x^2+7)) e^{\sqrt{\sin(5 \ln(x^2+7))}}}{(x^2+7) \sqrt{\sin(5 \ln(x^2+7))}}$$

$$(iv) \quad y = \sec x \sqrt{\tan x}$$

$$\frac{dy}{dx} = \sec x \frac{d}{dx} \sqrt{\tan x} + \sqrt{\tan x} \frac{d}{dx} (\sec x)$$

$$= \sec x \frac{1}{2\sqrt{\tan x}} \sec^2 x + \sqrt{\tan x} \cdot \sec x \cdot \tan x$$

$$= \left(\frac{\sec^3 x + 2 \sec x \tan^2 x}{2 \sqrt{\tan x}} \right)$$

$$(v) \quad y = \exp \left(\cos^3 (\tan x^3)^2 \right)$$

$$\frac{dy}{dx} = \exp \left(\cos^3 (\tan x^3)^2 \right) \cdot \frac{d}{dx} \left(\cos^3 (\tan x^3)^2 \right)$$

$$= y \cdot 3 \cos^2 (\tan x^3)^2 \cdot \frac{d}{dx} \cos (\tan x^3)^2$$

$$= 3y \cos^2 (\tan x^3)^2 (-\sin (\tan x^3)^2) \cdot \frac{d}{dx} (\tan x^3)^2$$

$$= -3y \cos^2 (\tan x^3)^2 \sin (\tan x^3)^2 (2 \tan(x^3) \cdot \sec^2 x^3) \cdot 3x^2$$

$$= -18y x^2 \cos^2 (\tan x^3)^2 \cdot \sin (\tan x^3)^2 \tan (x^3) \sec^2 x^3.$$

Illustration :

If $f(x) = (1+x)(3+x^2)^{1/2}(9+x^3)^{1/3}$ then $f'(-1)$ is equal to

- (A) 0 (B) $2\sqrt{2}$ (C) 4 (D) 6

Sol. $f(x) = (1+x)(3+x^2)^{1/2}(9+x^3)^{1/3}$

$$f'(x) = (3+x^2)^{1/2}(9+x^3)^{1/3} + (1+x) \frac{1}{2}(3+x^2)^{-1/2} \cdot 2x(9+x^3)^{1/3}$$

$$+ (1+x)(3+x^2)^{1/2} \left(\frac{1}{3}(9+x^3)^{-2/3} \cdot 3x^2 \right)$$

$$f'(-1) = (3+1)^{1/2}(9-1)^{1/3} + 0 + 0$$

$$f'(-1) = 2 \cdot 2 = 4$$

Practice Problem

Q.1 Differentiate the following functions with respect to x using first principle

- (i) $\sqrt{\sin x}$ (ii) $\cos^3 x$ (iii) $\tan^{-1} x$ (iv) $\log_e x$

Q.2 Find the derivative of the following functions with respect to x

- (i) $\sin(x^2)$ (ii) $\sin(\cos(x^2))$ (iii) $\sin(x^2+5)$ (iv) $\cos(\sin x)$
 (v) $\sin(ax+b)$ (vi) $\sec(\tan(\sqrt{x}))$ (vii) $\frac{\sin(ax+b)}{\cos(cx+d)}$ (viii) $\cos x^3 \cdot \sin^2(x^5)$
 (ix) $2\sqrt{\cos(x^2)}$ (x) $\cos(\sqrt{x})$

Q.3 Find the derivative of the following functions with respect to x

- (i) e^{-x} (ii) $\sin(\log x), x > 0$ (iii) $e^{\cos x}$
 (iv) $\frac{e^x}{\sin x}$ (v) $\sin(\tan^{-1} e^{-x})$ (vi) $\log(\cos e^x)$
 (vii) $e^x + e^{x^2} + \dots + e^{x^5}$ (viii) $\sqrt{e^{\sqrt{x}}}, x > 0$ (ix) $\log(\log x), x > 1$
 (x) $\frac{\cos x}{\log x}, x > 0$ (xi) $\cos(\log x + e^x), x > 0$

Q.4 Find the derivative of the following functions with respect to x

- (i) $x^2 + 3x + 2$ (ii) x^{20} (iii) $x \cdot \cos x$ (iv) $\log x$
 (v) $x^3 \log x$ (vi) $e^x \sin 5x$ (vii) $e^{6x} \cos 3x$ (viii) $\tan^{-1} x$
 (ix) $\log(\log x)$ (x) $\sin(\log x)$ (xi) $\sqrt{3x+2} + \frac{1}{\sqrt{2x^2+4}}$ (xii) $\log_7(\log x)$

Answer key

- Q.1** (i) $\frac{\cos x}{2\sqrt{\sin x}}$ (ii) $-3 \cos^2 x \sin x$ (iii) $\frac{1}{1+x^2}$ (iv) $\frac{1}{x}$
- Q.2** (i) $2x \cos x^2$ (ii) $-\cos(\cos x^2) \sin(x^2) \cdot 2x$ (iii) $2x \cos(x^2 + 5)$
 (iv) $-\sin(\sin x) \cos x$ (v) $a \cos(ax + b)$ (vi) $\frac{\sec(\tan \sqrt{x}) \tan(\tan \sqrt{x}) \sec^2 \sqrt{x}}{2\sqrt{x}}$
 (vii) $a \cos(ax + b) \sec(cx + d) + c \sin(ax + b) \tan(cx + d) \sec(cx + d)$
 (viii) $-3x^2 \sin x^3 \sin^2 x^5$ (ix) $\frac{-2\sqrt{2}x}{\sin x^2 \sqrt{\sin 2x^2}}$ (x) $\frac{-\sin \sqrt{x}}{2\sqrt{x}}$
- Q.3** (i) $-e^{-x}$ (ii) $\frac{\cos(\log x)}{x}$ (iii) $-e^{\cos x} \sin x$ (iv) $\frac{e^x(\sin x - \cos x)}{\sin^2 x}$
 (v) $\frac{-e^{-x} \cos(\tan^{-1} e^{-x})}{1+e^{-2x}}$ (vi) $-e^x \tan e^x, e^x \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{N}$
 (vii) $e^x + 2xe^{x^2} + \dots + 5x^4 e^{x^5}$ (viii) $\frac{e^{\sqrt{x}}}{4\sqrt{x}e^{\sqrt{x}}}, x > 0$ (ix) $\frac{1}{x \log x}$
 (x) $\frac{(x \sin x \log x + \cot x)}{x(\log x)^2}$ (xi) $-\left(\frac{1}{x} + e^x\right) \sin(\log x + e^x), x > 0$
- Q.4** (i) $2x + 3$ (ii) $20x^{19}$ (iii) $\cos x - x \sin$ (iv) $\frac{1}{x}$ (v) $x^2 + 3x^2 \log x$
 (vi) $e^x (\sin 5x + 5 \cos 5x)$ (vii) $e^{6x} (6 \cos 3x - 3 \sin 3x)$ (viii) $\frac{1}{x^2 + 1}$
 (ix) $\frac{1}{x \log x}$ (x) $\frac{\cos(\log x)}{x}$ (xi) $\frac{3}{2\sqrt{(3x+2)}} - \frac{2x}{(2x^2+4)^{3/2}}$ (xii) $\frac{\log_7 e}{x \log x}$

T-6 LOGARITHMIC DIFFERENTIATION :

To find the derivative of:

- (i) a function which is the product or quotient of a number of functions

$$y = f_1(x) f_2(x) f_3(x) \dots \quad \text{or} \quad y = \frac{f_1(x) f_2(x) f_3(x) \dots}{g_1(x) g_2(x) g_3(x) \dots}$$

- (ii) a function of the form $[f(x)]^{g(x)}$ where f & g are both derivable, it will be found convenient to take the logarithm of the function first & then differentiate

OR

$$y = (f(x))^{g(x)} = e^{g(x) \ln(f(x))} \text{ and then differentiate.}$$

Illustration :

If $y = \sin x \cdot \sin 2x \cdot \sin 3x \dots \sin nx$, find y' .

Sol. $y = \sin x \sin 2x \sin 3x \dots \sin nx$
 $\ln y = \ln (\sin x) + \ln (\sin 2x) + \ln (\sin 3x) + \dots + \ln (\sin nx)$
 $\frac{1}{y} \frac{dy}{dx} = \frac{\cos x}{\sin x} + \frac{2 \cos 2x}{\sin 2x} + \frac{3 \cos 3x}{\sin 3x} + \dots + \frac{n \cos nx}{\sin nx}$
 $y' = y (\cot x + 2 \cot 2x + 3 \cot 3x + \dots + n \cot nx).$

Illustration :

If $f(x) = (x+1)(x+2)(x+3) \dots (x+n)$ then $f'(0)$ is

(A) $n!$ (B) $\frac{n(n+1)}{2}$ (C) $(n!)(\ln n!)$ (D) $n! \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n} \right)$

Sol. $f(x) = (x+1)(x+2)(x+3) \dots (x+n)$
 $\ln f(x) = \ln (x+1) + \ln (x+2) + \ln (x+3) + \dots + \ln (x+n)$
 $\frac{f'(x)}{f(x)} = \frac{1}{(x+1)} + \frac{1}{(x+2)} + \frac{1}{(x+3)} + \dots + \frac{1}{(x+n)}$
 $f(0) = n!$
 $f'(0) = n! \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right).$

Illustration :

If $f(x) = \prod_{n=1}^{100} (x-n)^{n(101-n)}$ then find $\frac{f(101)}{f'(101)}$.

Sol. $f(x) = \prod_{n=1}^{100} (x-n)^{n(101-n)}$
 $f(x) = (x-1)^{1(101-1)} (x-2)^{2(101-2)} (x-3)^{3(101-3)} \dots (x-100)^{100(101-100)}$
 $\ln f(x) = 100 \ln (x-1) + 2(101-2) \ln (x-2) + 3(101-3) \ln (x-3) + \dots + 100 \ln (x-100)$
 $\frac{f'(x)}{f(x)} = \frac{100}{(x-1)} + \frac{2(101-2)}{(x-2)} + \frac{3(101-3)}{(x-3)} + \dots + \frac{100}{(x-100)}$
 $\left. \frac{f'(x)}{f(x)} \right|_{x=101} = 1 + 2 + 3 + \dots + 100 = (100) \frac{(100+1)}{2} = (50)(101) = 5050$
 So $\frac{f(101)}{f'(101)} = \frac{1}{5050}.$

Illustration :

Find the derivative of the following functions

$$(i) y = (\sin x) \left(e^{\sqrt{\sin x}} \right) (\ln x) (x^x); \quad (ii) y = (x^{\ln x}) (\sec x)^{3x}; \quad (iii) y = \frac{(\ln x)^x}{2^{3x+1}}$$

Sol.

(i) $y = (\sin x) \left(e^{\sqrt{\sin x}} \right) (\ln x) (x^x)$ take logarithm on both sides

$$\ln y = \ln (\sin x) + \sqrt{\sin x} + \ln (\ln x) + x \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{\cos x}{\sin x} + \frac{(\cos x)}{2\sqrt{\sin x}} + \frac{1}{(\ln x)} \cdot \frac{1}{x} + \ln x + 1$$

$$\frac{dy}{dx} = y \left(\cot x + \frac{1}{2} \frac{\cos x}{\sqrt{\sin x}} + \frac{1}{x \ln x} + \ln x + 1 \right)$$

(ii) $y = x^{\ln x} (\sec x)^{3x}$

$$\ln y = (\ln x)^2 + 3x \ln (\sec x)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{2 \ln x}{x} + 3 \ln (\sec x) + \frac{3x}{(\sec x)} (\sec x \tan x)$$

$$\frac{dy}{dx} = y \left(\frac{2 \ln x}{x} + 3 \ln (\sec x) + 3x \tan x \right).$$

(iii) $y = \frac{(\ln x)^x}{2^{3x+1}}$

$$\ln y = x \ln (\ln x) - (3x + 1) \ln 2$$

$$\frac{1}{y} \frac{dy}{dx} = \ln (\ln x) + \frac{x}{(\ln x)} \cdot \frac{1}{x} - 3 \ln 2$$

$$\frac{dy}{dx} = y \left(\ln (\ln x) + \frac{1}{(\ln x)} - 3 \ln 2 \right).$$

Illustration :

If $y = (\sin x)^{\ln x} \operatorname{cosec} (e^x (a + bx))$ and $a + b = \frac{\pi}{2e}$ then the value of $\frac{dy}{dx}$ at $x = 1$ is

(A) $(\sin 1) \ln \sin(1)$ (B) 0 (C) $\ln \sin(1)$ (D) $1 + \ln(\sin 1)$

Sol. $y = (\sin x)^{\ln x} \operatorname{cosec} (e^x (a + bx))$

$$\ln y = (\ln x) \ln (\sin x) - \ln (\sin (e^x (a + bx)))$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{x} \ln (\sin x) + (\ln x) \cot x - \frac{\cos(e^x (a + bx)) (ae^x + bxe^x + be^x)}{\sin(e^x (a + bx))}$$

$$\left. \frac{1}{y} \frac{dy}{dx} \right|_{at x=1} = \ln(\sin 1) + 0 - \frac{\cos\left(\frac{\pi}{2}\right)(a+2b)e}{\sin\left(\frac{\pi}{2}\right)} = \ln(\sin 1) + 0 - 0$$

$$\therefore a + b = \frac{\pi}{2e} \quad \therefore y = 1 \text{ at } x = 1$$

$$\text{so } \left. \frac{dy}{dx} \right|_{at x=1} = \ln(\sin 1) y \Big|_{at x=1} = \ln(\sin 1) \cdot 1 = \ln(\sin 1)$$

Illustration :

$$\text{If } y = 2^{\log_2 x^{2x}} + \left(\tan \frac{\pi x}{4} \right)^{\frac{4}{\pi x}} \text{ then } \left. \frac{dy}{dx} \right|_{x=1} \text{ is :}$$

- (A) 4 (B) 5/2 (C) 3 (D) not defined

$$\text{Sol. } y = 2^{\log_2 x^{2x}} + \left(\tan \frac{\pi x}{4} \right)^{\frac{4}{\pi x}}$$

$$y = x^{2x} + \left(\tan \frac{\pi x}{4} \right)^{\frac{4}{\pi x}}$$

$$\text{Let } u = x^{2x} \text{ so } \ln u = 2x \ln x$$

$$\frac{1}{u} \frac{du}{dx} = 2(\ln x + 1) \Rightarrow \left. \frac{du}{dx} \right|_{x=1} = (1)(2) = 2$$

$$\text{Let } v = \left(\tan \frac{\pi x}{4} \right)^{\frac{4}{\pi x}}$$

$$\ln v = \frac{4}{\pi x} \ln \left(\tan \left(\frac{\pi x}{4} \right) \right)$$

$$\frac{1}{v} \frac{dv}{dx} = -\frac{4}{\pi x^2} \ln \left(\tan \left(\frac{\pi x}{4} \right) \right) + \frac{1}{x} \frac{\sec^2 \left(\frac{\pi}{4} x \right)}{\tan \left(\frac{\pi}{4} x \right)}$$

$$\left. \frac{dv}{dx} \right|_{at x=1} = 0 + 2$$

$$\text{So } \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx} = 2 + 2 = 4.$$

Illustration :

Find $\frac{dy}{dx}$ for (i) $y = (\sin x)^{\log x}$ (ii) $y = x^{\tan x} + (\sin x)^{\cos x}$

Sol.

(i) Given $y = (\sin x)^{\log x}$.

Then, $y = e^{\log x \log \sin x}$

Diff. both sides w.r.t. x , we get

$$\frac{dy}{dx} = e^{\log x \log \sin x} \frac{d}{dx} \{\log x \log \sin x\}$$

$$\Rightarrow \frac{dy}{dx} = (\sin x)^{\log x} \left\{ \log \sin x \frac{d}{dx} (\log x) + \log x \frac{d}{dx} (\log \sin x) \right\}$$

$$\Rightarrow \frac{dy}{dx} = (\sin x)^{\log x} \left\{ \frac{\log \sin x}{x} + \log x \frac{1}{\sin x} \cos x \right\}$$

$$\Rightarrow \frac{dy}{dx} = (\sin x)^{\log x} \left\{ \frac{\log \sin x}{x} + \cot x \log x \right\}.$$

(ii) $y = x^{\tan x} + (\sin x)^{\cos x}$

Let $u = x^{\tan x}$, $v = (\sin x)^{\cos x}$

$$\therefore \ln u = \tan x \ln x$$

$$\therefore \frac{d}{dx} (\ln u) = \frac{d}{dx} (\tan x \cdot \ln x)$$

$$\frac{1}{u} \cdot \frac{du}{dx} = \tan x \left(\frac{1}{x} \right) + \sec^2 x \cdot \ln x$$

$$\frac{du}{dx} = x^{\tan x} \left(\frac{\tan x}{x} + \sec^2 x \cdot \ln x \right)$$

$$v = (\sin x)^{\cos x}$$

$$\ln v = \cos x \cdot \ln \sin x$$

$$\Rightarrow \frac{d}{dx} \ln v = \frac{d}{dx} (\cos x \ln \sin x)$$

$$\frac{1}{v} \frac{dv}{dx} = \left(\cos x \cdot \frac{\cos x}{\sin x} \right) + \ln \sin x \cdot (-\sin x)$$

$$\frac{dv}{dx} = (\sin x)^{\cos x} \left(\frac{\cos^2 x}{\sin x} - \sin x \ln (\sin x) \right) \quad \dots\dots(ii)$$

$$y = u + v$$

$$\therefore \frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}.$$

From (i) and (ii)

$$\Rightarrow \frac{dy}{dx} = x^{\tan x} \left(\frac{\tan x}{x} + \sec^2 x \cdot \ln x \right) + (\sin x)^{\cos x} \left(\frac{\cos^2 x}{\sin x} - \sin x \cdot \ln (\sin x) \right).$$

Illustration :

Find the derivative of $\frac{\sqrt{x}(x+4)^{3/2}}{(4x-3)^{4/3}}$ w.r.t. x ,

Sol. Let $y = \frac{\sqrt{x}(x+4)^{3/2}}{(4x-3)^{4/3}}$

Taking log of both sides, we get

$$\log y = \frac{1}{2} \log x + \frac{3}{2} \log (x+4) - \frac{4}{3} \log (4x-3)$$

Diff. both sides w.r.t. x , we get

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2x} + \frac{3}{2x+4} \times 1 - \frac{4}{3} \times \frac{1}{4x-3} \times 4$$

$$\Rightarrow \frac{dy}{dx} = y \left\{ \frac{1}{2x} + \frac{3}{2(x+4)} - \frac{16}{3(4x-3)} \right\}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{x}(x+4)^{3/2}}{(4x-3)^{4/3}} \left\{ \frac{1}{2x} + \frac{3}{2(x+4)} - \frac{16}{3(4x-3)} \right\}$$

Illustration :

Find $\frac{dy}{dx}$ for $y = x^x$.

Sol. $y = x^x$

$$\ln y = x \ln x \quad \Rightarrow \quad \frac{d}{dx} (\ln y) = \frac{d}{dx} (x \ln x)$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = x \cdot \left(\frac{1}{x} \right) + \ln x = 1 + \ln x \quad \Rightarrow \quad \frac{dy}{dx} = x^x (1 + \ln x).$$

Illustration :

If $y^x = x^y$, then find $\frac{dy}{dx}$.

Sol. $y^x = x^y$

$$x \ln y = y \ln x$$

Differentiating w.r.t. x .

$$\frac{d}{dx} (x \ln y) = \frac{d}{dx} (y \ln x)$$

$$x \cdot \left(\frac{1}{y} \cdot \frac{dy}{dx} \right) + \ln y = y \cdot \frac{1}{x} + \ln x \cdot \frac{dy}{dx} \quad \Rightarrow \quad \frac{dy}{dx} \left(\frac{x}{y} - \ln x \right) = \frac{y}{x} - \ln y$$

$$\frac{dy}{dx} = \frac{\frac{x}{y} - \ln y}{\frac{x}{y} - \ln x} = \frac{y(y - x \ln y)}{x(x - y \ln x)}$$

T-7 IMPLICIT FUNCTIONS :

If the variable x and y are connected by a relation of the form $f(x, y) = 0$ and it is not possible to express y as a function of x in the form $y = \phi(x)$, then such functions are said to be implicit functions.

For example,

- (i) $x^2 + xy + y^3 = 1$ (ii) $x + y + \sin(xy) = 1$ (iii) $x^y + y^x = 1$
(iv) $16^{x^2+y} + 16^{x+y^2} = 1$

DERIVATIVE OF IMPLICIT FUNCTION :

- (i) In order to find dy/dx , in the case of implicit functions, we differentiate each term w.r.t. x regarding y as a functions of x & then collect terms in dy/dx together on one side to finally find dy/dx .
- (ii) Corresponding to every curve represented by an implicit equation, there exist one or more explicit functions representing that equation. It can be shown that dy/dx at any point on the curve remains the same whether the process of differentiation is done explicitly or implicitly.

$$\frac{d}{dx} f(y) = \frac{d}{dy} f(y) \cdot \frac{dy}{dx} = f'(y) \cdot \frac{dy}{dx}$$

For example :

$$(i) \quad \frac{d}{dx} (\sin y) = \frac{d}{dy} (\sin y) \cdot \frac{dy}{dx} = \cos y \left(\frac{dy}{dx} \right) \quad (ii) \quad \frac{d}{dx} (y^3) = \frac{d}{dy} (y^3) \cdot \frac{dy}{dx} = 3y^2 \frac{dy}{dx}$$

A DIRECT FORMULA FOR IMPLICIT FUNCTIONS :

Let $f(x, y) = 0$. Take all the terms of left side and put left side equal to $f(x, y)$.

$$\text{Then } \frac{dy}{dx} = - \frac{\text{diff. of } f \text{ w.r.t. } x \text{ keeping } y \text{ as constant}}{\text{diff. of } f \text{ w.r.t. } y \text{ keeping } x \text{ as constant}}$$

Illustration :

If $x^2 + 2xy + y^3 = 4$, find $\frac{dy}{dx}$.

Sol. We have $x^2 + 2xy + y^3 = 4$

Diff. both sides w.r.t. x ,

$$\Rightarrow \frac{d}{dx} (x^2) + 2 \frac{d}{dx} (xy) + \frac{d}{dx} (y^3) = \frac{d}{dx} (4)$$

$$\Rightarrow 2x + 2 \left(x \frac{dy}{dx} + y \cdot 1 \right) + 3y^2 \frac{dy}{dx} = 0 \quad \Rightarrow \frac{dy}{dx} = - \frac{2(x+y)}{(2x+3y^2)}$$

Alternative Method :

$$\frac{dy}{dx} = - \frac{\text{diff. of } f \text{ w.r.t. } x \text{ keeping } y \text{ as constant}}{\text{diff. of } f \text{ w.r.t. } y \text{ keeping } x \text{ as constant}} = - \frac{2x+2y}{2x+3y^2}$$

Illustration :

If $y = x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots}}}$, prove that $\frac{dy}{dx} = \frac{y}{2y-x}$.

Sol. We have, $y = x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots}}}$,

$$\Rightarrow y = x + \frac{1}{y} \quad \Rightarrow y^2 = xy + 1$$

$$\Rightarrow 2y \frac{dy}{dx} = y + x \frac{dy}{dx} + 0 \quad [\text{Diff. both sides with respect to } x]$$

$$\Rightarrow \frac{dy}{dx} (2y - x) = y \quad \Rightarrow \frac{dy}{dx} = \frac{y}{2y-x}.$$

Illustration :

If $\sqrt{x} + \sqrt{y} = 4$, then find $\frac{dx}{dy}$ at $y = 1$.

Sol. Diff. both sides of the given equation w.r.t. y , we get

$$\frac{1}{2\sqrt{x}} \frac{dx}{dy} + \frac{1}{2\sqrt{y}} = 0 \Rightarrow \frac{dx}{dy} = -\frac{\sqrt{x}}{\sqrt{y}} = \frac{\sqrt{y}-4}{\sqrt{y}} \Rightarrow \left[\frac{dx}{dy} \right]_{y=1} = \frac{1-4}{1} = -3.$$

Illustration :

If $x^y = e^{x-y}$ then prove that $\frac{dy}{dx} = \frac{\ln x}{(1+\ln x)^2}$.

Sol. $x^y = e^{x-y}$

$y \ln x = (x-y) \dots\dots(i)$

now differentiating with respect to x .

$$\frac{y}{x} + \ln x \cdot \frac{dy}{dx} = 1 - \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1-\frac{y}{x}}{1+\ln x} = \frac{x-y}{x(1+\ln x)} = \frac{y \ln x}{x(1+\ln x)} \quad \left(\text{From (i), } y = \frac{x}{1+\ln x} \right)$$

$$\therefore \frac{dy}{dx} = \frac{\ln x}{(1+\ln x)^2}.$$

Illustration :

If $\sin y = x \sin(a + y)$ then prove that $\frac{dy}{dx} = \frac{\sin^2(a + y)}{\sin a}$.

Sol. $\sin y = x \sin(a + y)$ differentiating both sides w.r.t. x

$$\cos y \cdot \frac{dy}{dx} = \sin(a + y) + x \cos(a + y) \cdot \left(0 + \frac{dy}{dx}\right)$$

$$\Rightarrow \frac{dy}{dx} (\cos y - x \cos(a + y)) = \sin(a + y)$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sin(a + y)}{\cos y - x \cos(a + y)} = \frac{\sin(a + y)}{\cos y - \frac{\sin y}{\sin(a + y)} \cos(a + y)}$$

$$= \frac{\sin^2(a + y)}{\sin(a + y) \cos y - \sin y \cos(a + y)} = \frac{\sin^2(a + y)}{\sin a}$$

Illustration :

If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, then prove that $\frac{dy}{dx} = -\frac{1}{(1+x)^2}$

Sol. $x\sqrt{1+y} + y\sqrt{1+x} = 0$

$$\Rightarrow x^2(1+y) = y^2(1+x) \quad \Rightarrow (x^2 - y^2) + x^2y - y^2x = 0$$

$$\Rightarrow (x-y)(x+y+xy) = 0 \quad \Rightarrow x-y = 0 \text{ or } x+y+xy = 0$$

$$\therefore y = x \text{ or } \frac{-x}{x+1} \quad \Rightarrow \frac{dy}{dx} = 1 \text{ or } \frac{-1}{(x+1)^2}$$

Illustration :

If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}}$ find $\frac{dy}{dx}$ ($\sin x > 0$).

Sol. $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \infty}}} \Rightarrow y = \sqrt{\sin x + y}$

$$\Rightarrow y^2 - y = \sin x \Rightarrow 2y \frac{dy}{dx} - \frac{dy}{dx} = \cos x \Rightarrow \frac{dy}{dx} = \left(\frac{\cos x}{2y-1} \right)$$

Illustration :

If $ax^2 + 2hxy + by^2 = 0$ then prove that $\frac{dy}{dx} = -\frac{ax+hy}{hx+by} = \frac{y}{x}$

Sol. $ax^2 + 2hxy + by^2 = 0$ (1)

differentiating w.r.t. x .

$$2ax + 2h \left(x \frac{dy}{dx} + y \right) + 2by \frac{dy}{dx} = 0 \quad \Rightarrow (hx + by) \frac{dy}{dx} = -(ax + hy)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{(ax+hy)}{(hx+by)} \dots\dots\dots(2)$$

From (1) $ax^2 + 2hxy + by^2 = 0$

$$\Rightarrow x(ax+hy) + y(hx+by) = 0$$

$$\Rightarrow \frac{ax+hy}{hx+by} = \frac{-y}{x}$$

From (ii) $\frac{dy}{dx} = \frac{-(ax+hy)}{(hx+by)} = \frac{y}{x}$

Illustration :

A curve is described by the relation $\ln(x+y) = xe^y$. Find the tangent to the curve at $(0, 1)$.

Sol. $\ln(x+y) = xe^y$
differentiating w.r.t. to x

$$\frac{1}{(x+y)} \left(1 + \frac{dy}{dx} \right) = xe^y \frac{dy}{dx} + e^y \text{ at point } (0, 1),$$

$$\frac{1}{(0+1)} \left(1 + \frac{dy}{dx} \right) = 0 + e^1$$

$$\Rightarrow \frac{dy}{dx} \Big|_{x=0, y=1} = e - 1$$

Equation of tangent $y - 1 = (e - 1)(x - 0) \Rightarrow (1 - e)x + y = 1$.

Illustration :

(i) $y = x^{x^{x^{\dots\infty}}}$ (ii) $y = (\ln x)^{(\ln x)^{(\ln x)^{\dots\infty}}}$

Sol.

(i) $y = x^{x^{x^{\dots\infty}}}$

$$\Rightarrow y = x^y \Rightarrow \ln y = y \ln x \Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \frac{y}{x} + \ln x \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \left(\frac{1}{y} - \ln x \right) = \frac{y}{x} \Rightarrow \frac{dy}{dx} = \frac{\frac{y}{x}}{(1 - y \ln x)} = \frac{y^2}{x(1 - y \ln x)}$$

(ii) $y = (\ln x)^y$

$$\Rightarrow \ln y = y \ln(\ln x) \Rightarrow \frac{1}{y} \cdot \frac{dy}{dx} = \frac{y}{x \ln x} + \ln \ln x \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{1}{y} - \ln \ln x \right) = \frac{y}{x \ln x} \Rightarrow \frac{dy}{dx} \frac{\frac{y}{x \ln x}}{(1 - y \ln \ln x)} = \frac{y^2}{x \ln x (1 - y \ln \ln x)}$$

Illustration :

If $y^5 + xy^2 + x^3 = 4x + 3$, then find $\frac{dy}{dx}$ at $(2, 1)$

Sol. $y^5 + xy^2 + x^3 = 4x + 3$

differentiating w.r.t. x

$$5y^4 \frac{dy}{dx} + \left(x \cdot 2y \frac{dy}{dx} + y^2 \right) + 3x^2 = 4 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{4 - 3x^2 - y^2}{(5y^4 + 2xy)}$$

$$\left. \frac{dy}{dx} \right|_{x=2, y=1} = \frac{4 - 3(2)^2 - (1)^2}{5(1)^4 + 2(2)(1)} = \frac{4 - 12 - 1}{5 + 4} = \frac{-9}{9} = -1.$$

Illustration :

If $y = \sqrt{x \log_e x}$, then find $\frac{dy}{dx}$ at $x = e$.

Sol. $\frac{dy}{dx} = \frac{1}{2\sqrt{x \log_e x}} \cdot \frac{d}{dx} [x \log_e x] = \frac{1}{2\sqrt{x \log_e x}} \left[x \cdot \frac{1}{x} + 1 \cdot \log_e x \right]$

$$\Rightarrow \left[\frac{dy}{dx} \right]_{x=e} = \frac{1}{2\sqrt{e \cdot 1}} (1 + 1) = \frac{1}{\sqrt{e}} \quad (\because \log_e e = 1)$$

Practice Problem

Q.1 Find the derivative of the following functions with respect to x :

(i) $\sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}}$

(ii) $x^{\sin x}, x > 0$

(iii) $y^x + x^y + x^x = a^b$

(iv) $\cos x \cdot \cos 2x \cdot \cos 3x$ (v) $\sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)(x-5)}}$

(vi) $(\log x)^{\cos x}$

(vii) $x^x - 2^{\sin x}$

(viii) $(x+3)^2 \cdot (x+4)^3 \cdot (x+5)^4$

(ix) $\left(x + \frac{1}{x}\right)^x + x^{\left(1 + \frac{1}{x}\right)}$

(x) $(\log x)^x + x^{\log x}$

(xi) $x^{x \cos x} + \frac{x^2 + 1}{x^2 - 1}$

(xii) $x^{\sin x} + (\sin x)^{\cos x}$

Q.2 Find $\frac{dy}{dx}$ of the following functions :

(i) $(x \cos x)^x + (x \sin x)^{\frac{1}{x}}$

(ii) $x^y + y^x = 1$

(iii) $y^x = x^y$

(iv) $(\cos x)^y = (\cos y)^x$

(v) $xy = e^{(x-y)}$

(vi) $y = (x^2 - 5x + 8)(x^3 + 7x + 9)$

(vii) $y = (1+x)(1+x^2)(1+x^4)(1+x^8)$

Q.3 Find $\frac{dy}{dx}$ of the following functions :

- | | |
|--------------------------------|-------------------------------------|
| (i) $2x + 3y = \sin x$ | (ii) $2x + 3y = \sin y$ |
| (iii) $ax + by^2 = \cos y$ | (iv) $xy + y^2 = \tan x + y$ |
| (v) $x^2 + xy + y^2 = 100$ | (vi) $x^3 + x^2y + xy^2 + y^3 = 81$ |
| (vii) $\sin^2 y + \cos xy = k$ | (viii) $x^2 + xy + \cos^2 y = 1$ |
| (ix) $y + \sin y = \cos x$ | |

Q.4 Find the derivative of the following functions with respect to x :

- | | |
|--|--|
| (i) $(\sin x)^{\sin x}, 0 < x < \pi.$ | (ii) $(\sin x - \cos x)^{(\sin x - \cos x)}, \frac{\pi}{4} < x < \frac{3\pi}{4}$ |
| (iii) $x^x + x^a + a^x + a^a$, for some fixed $a > 0$ and $x > 0$ | |
| (iv) $x^{x^2-3} + (x-3)^{x^2}$, for $x > 3$ | (v) $\log \left\{ e^{x \left(\frac{x-2}{x+2} \right)^{\frac{3}{4}}} \right\}$ |

Answer key

- Q.1**
- | | |
|--------|---|
| (i) | $\frac{1}{2} \sqrt{\frac{(x-3)(x^2+4)}{3x^2+4x+5}} \left[\frac{1}{x-3} + \frac{2x}{x^2+4} - \frac{6x+4}{3x^2+4x+5} \right]$ |
| (ii) | $x^{\sin x-1} \sin x + x^{\sin x} \cdot \cos x \log x$ |
| (iii) | $\frac{-[y^x \log y + y \cdot x^{y-1} + x^x(1 + \log x)]}{x \cdot y^{x-1} + x^y \log x}$ |
| (iv) | $-\cos x \cos 2x \cos 3x [\tan x + 2 \tan 2x + 3 \tan 3x]$ |
| (v) | $\frac{1}{2} \sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)(x-5)}} \left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4} - \frac{1}{x-5} \right]$ |
| (vi) | $(\log x)^{\cos x} \left[\frac{\cos x}{x \log x} - \sin x \log \log x \right]$ |
| (vii) | $x^x (1 + \log x) - 2^{\sin x} \cos x \log 2$ |
| (viii) | $(x+3)(x+4)^2(x+5)^3(9x^2+70x+133)$ |
| (ix) | $\left(x + \frac{1}{x} \right)^x \left[\frac{x^2-1}{x^2+1} + \log \left(x + \frac{1}{x} \right) \right] + x^{1+\frac{1}{x}} \left(\frac{x+1-\log x}{x^2} \right)$ |
| (x) | $(\log x)^{x-1} [1 + \log x \cdot \log \log x] + 2x^{\log x-1} \log x$ |
| (xi) | $x^x \cos x [\cos x (1 + \log x) - x \sin x \log x] - \frac{4x}{(x^2-1)^2}$ |
| (xii) | $x^{\sin x} \left[\frac{\sin x}{x} + \cos x \log x \right] + (\sin x)^{\cos x} [\cos x \cot x - \sin x \log \sin x]$ |
-

- Q.2** (i) $(x \cos x)^x [1 - x \tan x + \log(x \cos x)] + (x \sin x)^{1/x} \left[\frac{x \cot x + 1 - \log(x \sin x)}{x^2} \right]$
- (ii) $-\left(\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}} \right)$ (iii) $\frac{y}{x} \left(\frac{y - x \log y}{x - y \log x} \right)$ (iv) $\frac{y \tan x + \log \cos y}{x \tan y + \log \cos x}$
- (v) $\frac{y(x-1)}{x(y+1)}$ (vi) $5x^4 - 20x^3 + 45x^2 - 52x + 11$
- (vii) $(1+x)(1+x^2)(1+x^4)(1+x^8) \left[\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \frac{8x^7}{1+x^8} \right]$
- Q.3** (i) $\frac{\cos x - 2}{3}$ (ii) $\frac{2}{\cos y - 3}$ (iii) $\frac{-9}{2by + \sin y}$ (iv) $\frac{\sec^2 x - y}{x + 2y - 1}$
- (v) $-\frac{(2x+y)}{(x+2y)}$ (vi) $\frac{-(3x^2 + 2xy + y^2)}{(x^2 + 2xy + 3y^2)}$ (vii) $\frac{y \sin(xy)}{\sin 2y - x \sin xy}$ (viii) $\frac{2x+y}{\sin 2y + x}$
- (ix) $\frac{-\sin x}{1 + \cos y}$
- Q.4** (i) $(1 + \log \sin x) (\sin x)^{\sin x} \cos x$
- (ii) $(\sin x - \cos x)^{\sin - \cos x} (\cos x + \sin x) [1 + \log(\sin x - \cos x)], \sin x > \cos x$
- (iii) $x^x (1 + \log x) + ax^{a-1} + a^x \log a$
- (iv) $x^{x^2-3} \left[\frac{x^2-3}{x} + 2x \log x \right] + (x^2-3)^{x^2} \left[\frac{x^2}{x-3} + 2x \log(x-3) \right]$ (v) $\frac{x^2-1}{x^2-4}$

T-8 DERIVATIVE OF INVERSE FUNCTION :

Theorem :

Let the functions $f(x)$ and $g(x)$ be inverse of each other then

$$f(g(x)) = g(f(x)) = x$$

$$\therefore f'(g(x)) g'(x) = g'(f(x)) f'(x) = 1$$

$$\text{If } \frac{dy}{dx} \text{ exists and } \frac{dy}{dx} \neq 0, \text{ then } \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} \text{ or } \frac{dy}{dx} \cdot \frac{dx}{dy} = 1 \text{ or } \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} \left[\frac{dx}{dy} \neq 0 \right]$$

Illustration :

- (a) If $y = f(x) = x^3 + x^5$ and g is the inverse of f find $g'(2)$
 (b) Let $f(x) = \exp(x^3 + x^2 + x)$ for any real number x , and let g be the inverse function for f . The value of $g'(e^3)$ is

(A) $\frac{1}{6e^3}$ (B) $\frac{1}{6}$ (C) $\frac{1}{34e^{39}}$ (D) 6

- (c) If g is the inverse of f and $f'(x) = \frac{1}{1+x^n}$, prove that $g'(x) = 1 + (g(x))^n$

Sol.

- (a) $y = f(x) = x^3 + 5$
 Let $g(x)$ be the inverse of $f(x)$ i.e., $g(x) = f^{-1}(x)$
 $f(g(x)) = g(f(x)) = x$
 differentiating w.r.t. x
 $f'(g(x)) \cdot g'(x) = 1$

$$\Rightarrow g'(x) = \frac{1}{f'(g(x))} \quad \Rightarrow \quad g(2) = f^{-1}(2) = y \text{ (say)}$$

$$\Rightarrow f(y) = 2 \Rightarrow y^3 + y^5 = 2 \Rightarrow y = 1$$

$$\Rightarrow g'(2) = \frac{1}{f'(1)} = \frac{1}{(3x^2 + 5x^4)_{x=1}} = \frac{1}{8}$$

- (b) $f(g(x)) = x$

$$\Rightarrow f'(g(x)) g'(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))} \Rightarrow g'(e^3) = \frac{1}{f'(g(e^3))}$$

$$\Rightarrow e^{x^3+x^2+x} = e^3 \Rightarrow x = 1$$

$$\Rightarrow g'(e^3) = \frac{1}{f'(1)} = \frac{1}{\left[e^{x^3+x^2+x} (3x^2 + 2x + 1) \right]_{x=1}} = \frac{1}{e^3 (3+2+1)} = \frac{1}{6e^3}$$

- (c) $f(g(x)) = x$

$$\Rightarrow f'(g(x)) g'(x) = 1 \Rightarrow g'(x) = \frac{1}{f'(g(x))} = \frac{1}{1 + (g(x))^n}$$

$$\Rightarrow g'(x) = 1 + (g(x))^n$$

DIFFERENTIATION OF INVERSE TRIGONOMETRIC FUNCTIONS :

$$(i) \quad \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$(ii) \quad \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$$

$$(iii) \quad \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$(iv) \quad \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$(v) \quad \frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$$

$$(vi) \quad \frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}$$

Proof:

(i) Proof of derivative of $f(x) = \sin^{-1} x$:

Let $y = \sin^{-1} x$. Then, $x = \sin y$.

Differentiating both sides w.r.t. x , we get

$$1 = \cos y \frac{dy}{dx}$$

which implies that $\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\cos(\sin^{-1} x)}$

this is defined only for $\cos y \neq 0$, i.e., $\sin^{-1} x \neq -\frac{\pi}{2}, \frac{\pi}{2}$, i.e., $x \neq -1, 1$, i.e., $x \in (-1, 1)$

Recall that for $x \in (-1, 1)$, $\sin(\sin^{-1} x) = x$ and hence

$$\cos^2 y = 1 - (\sin y)^2 = 1 - (\sin(\sin^{-1} x))^2 = 1 - x^2$$

Also, since $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $\cos y$ is positive and hence $\cos y = \sqrt{1-x^2}$

Thus, for $x \in (-1, 1)$,

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}$$

(ii) Proof of derivative of $f(x) = \cos^{-1} x$

$$\frac{d}{dx}(\cos^{-1} x) = \frac{d}{dx}\left(\frac{\pi}{2} - \sin^{-1} x\right) \quad \left[\because \cos^{-1} x + \sin^{-1} x = \frac{\pi}{2}\right]$$

$$= 0 - \frac{d}{dx}(\sin^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

(iii) Proof of derivative of $f(x) = \tan^{-1} x$;

Let $y = \tan^{-1} x$. Then, $x = \tan y$,

Differentiating both sides w.r.t. x , we get

$$1 = \sec^2 y \frac{dy}{dx}$$

which implies that

$$\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1+\tan y} = \frac{1}{1+(\tan(\tan^{-1} x))^2} = \frac{1}{1+x^2}$$

(iv) Proof of derivative of $f(x) = \cot^{-1} x$.

$$\begin{aligned}\frac{d}{dx}(\cot^{-1} x) &= \frac{d}{dx}\left(\frac{\pi}{2} - \tan^{-1} x\right) = 0 - \frac{d}{dx}(\tan^{-1} x) \\ &= -\frac{1}{(1+x^2)}\end{aligned}$$

Note :- $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$

(v) Proof of derivative of $f(x) = \sec^{-1} x$;

Let $y = \sec^{-1} x$; $x = \sec y$

Differentiating both sides w.r.t. x , we get

$$1 = \sec y \tan y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{\sec y \tan y} = \frac{1}{|x| \sqrt{x^2 - 1}}$$

(vi) Proof of derivative of $f(x) = \operatorname{cosec}^{-1} x$.

Let $y = \operatorname{cosec}^{-1} x$; $x = \operatorname{cosec} y$.

Differentiating both sides w.r.t. x , we get

$$1 = -\operatorname{cosec} y \cot y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\frac{1}{\operatorname{cosec} y \cot y} = -\frac{1}{x \sqrt{x^2 - 1}}$$

Finding of the derivatives of other inverse trigonometric function is left exercise. The following table gives the derivatives of the remaining inverse trigonometric functions.

$f(x)$	$\sin^{-1} x$	$\cos^{-1} x$	$\tan^{-1} x$	$\cot^{-1} x$	$\sec^{-1} x$	$\operatorname{cosec}^{-1} x$
$f'(x)$	$\frac{1}{\sqrt{1-x^2}}$	$\frac{-1}{\sqrt{1-x^2}}$	$\frac{1}{(1+x^2)}$	$\frac{1}{1+x^2}$	$\frac{1}{ x \sqrt{x^2-1}}$	$\frac{-1}{ x \sqrt{x^2-1}}$
Domain of f'	$(-1, 1)$	$(-1, 1)$	$\mathbb{R}$	$\mathbb{R}$	$(-\infty, -1) \cup (1, \infty)$	$(-\infty, -1) \cup (1, \infty)$

Note : Some Standard Substitutions

Expressions

$$\sqrt{a^2 - x^2}$$

$$\sqrt{a^2 + x^2}$$

$$\sqrt{x^2 - a^2}$$

$$\sqrt{\frac{a+x}{a-x}} \text{ or } \sqrt{\frac{a-x}{a+x}}$$

Substitution

$$x = a \sin \theta \text{ or } a \cos \theta$$

$$x = a \tan \theta \text{ or } a \cot \theta$$

$$x = a \sec \theta \text{ or } a \operatorname{cosec} \theta$$

$$x = a \cos \theta \text{ or } a \cos 2\theta$$

Illustration :

Prove that derivative of $\sec^{-1}x$ is $\frac{1}{|x|\sqrt{x^2-1}}$ if $|x| > 1$, using first principle method.

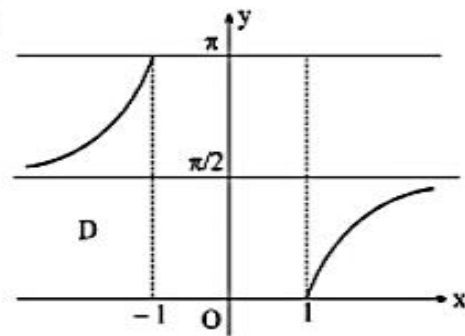
Sol. Let, $y = \sec^{-1}x$; $|x| \geq 1$; $y \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$

$$\sec y = x ; \sec(y + \Delta y) = x + \Delta x$$

$$\therefore \Delta x = \sec(y + \Delta y) - \sec y$$

$$\frac{\Delta x}{\Delta y} = \frac{\sec(y + \Delta y) - \sec y}{\Delta y}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta y \rightarrow 0} \frac{1}{\frac{\sec(y + \Delta y) - \sec y}{\Delta y}} = \frac{1}{\sec y \tan y}$$



if $x > 1$ then $y \in \left(0, \frac{\pi}{2}\right)$ (at $x = 1$, $y = 0$ and $\tan y = 0$ therefore $\frac{dy}{dx}$ does not exist)

$\Rightarrow$ sign of $\tan y$ is +ve

$$\therefore \frac{dy}{dx} = \frac{1}{\sec y \sqrt{\sec^2 y - 1}} = \frac{1}{x \sqrt{x^2 - 1}}, \text{ if } x > 1 \quad \dots(1)$$

and if $x < -1$, then $y \in \left(\frac{\pi}{2}, \pi\right)$ (at $x = -1$, $y = \pi$ and $\tan y = 0$ and $\frac{dy}{dx}$ does not exist)

hence sign of $\tan y$ is -ve

$$\therefore \frac{dy}{dx} = -\frac{1}{\sec y \sqrt{\sec^2 y - 1}} = -\frac{1}{x \sqrt{x^2 - 1}}, \text{ if } x < -1 \quad \dots(2)$$

from (1) and (2)

$$D(\sec^{-1}x) = \begin{cases} \frac{1}{|x|\sqrt{x^2-1}}, & \text{if } |x| > 1 \\ \text{does not exist, if } |x| = 1 \end{cases}$$

Illustration :

Find $\frac{dy}{dx}$ for $y = \tan^{-1} \left\{ \frac{1 - \cos x}{\sin x} \right\}$, where $-\pi < x < \pi$.

Sol. $y = \tan^{-1} \left\{ \frac{1 - \cos x}{\sin x} \right\} = \tan^{-1} \left\{ \frac{2 \sin \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}} \right\}$

$$= \tan^{-1} \left(\tan \frac{x}{2} \right) = \frac{x}{2} \quad \left(\because -\pi < x < \pi \Rightarrow -\frac{\pi}{2} < \frac{x}{2} < \frac{\pi}{2} \right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2}$$

Illustration :

$$D\left(\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)\right) = \text{---}$$

Sol. $\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \frac{\pi}{2}, & x > 0 \\ -\frac{\pi}{2}, & x < 0 \end{cases}$

$$D\left(\tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)\right) = 0 \quad \forall x \in \mathbb{R} - \{0\}.$$

Illustration :

If $y = \frac{\tan^{-1}x - \cot^{-1}x}{\tan^{-1}x + \cot^{-1}x}$, find $\frac{dy}{dx}\bigg|_{x=-1}$;

- (A) 0 (B) 1 (C) $2/\pi$ (D) -1

Sol. $y = \frac{\tan^{-1}x - \cot^{-1}x}{\tan^{-1}x + \cot^{-1}x} = \frac{2}{\pi}(\tan^{-1}x - \cot^{-1}x)$

$$\therefore \frac{dy}{dx} = \frac{2}{\pi} \left(\frac{1}{1+x} + \frac{1}{1+x^2} \right) = \frac{4}{\pi(1+x^2)}$$

$$\frac{dy}{dx}\bigg|_{x=-1} = \frac{4}{\pi(1+(-1)^2)} = \frac{2}{\pi}$$

Illustration :

If $y = \tan^{-1}\left(\frac{x}{1 \cdot 2 + x^2}\right) + \tan^{-1}\left(\frac{x}{2 \cdot 3 + x^2}\right) + \tan^{-1}\left(\frac{x}{3 \cdot 4 + x^2}\right) + \dots$ up to n times. Find $\frac{dy}{dx}$

expressing your answer in two terms and also find $\frac{dy}{dx}$ when $n \rightarrow \infty$.

Sol. $y = \tan^{-1}\left(\frac{x}{1 \cdot 2 + x^2}\right) + \tan^{-1}\left(\frac{x}{2 \cdot 3 + x^2}\right) + \tan^{-1}\left(\frac{x}{3 \cdot 4 + x^2}\right) + \dots + \tan^{-1}\left(\frac{x}{n(n+1) + x^2}\right)$

$$T_r = \tan^{-1}\left(\frac{x}{r(r+1) + x^2}\right) \Rightarrow T_r = \tan^{-1}\left(\frac{\frac{x}{r(r+1)}}{1 + \frac{x^2}{r(r+1)}}\right)$$

$$T_r = \tan^{-1}\left(\frac{\frac{x}{r} - \frac{x}{r+1}}{1 + \frac{x}{r} \cdot \frac{x}{r+1}}\right) = \tan^{-1}\left(\frac{x}{r}\right) - \tan^{-1}\left(\frac{x}{r+1}\right).$$

$$\begin{aligned}
 y = S_n &= \sum_{r=1}^n T_r = \sum_{r=1}^n \tan^{-1}\left(\frac{x}{r}\right) - \tan^{-1}\left(\frac{x}{r+1}\right) \\
 &= \left(\tan^{-1}(x) - \tan^{-1}\left(\frac{x}{2}\right) + \left(\tan^{-1}\left(\frac{x}{2}\right) - \tan^{-1}\left(\frac{x}{3}\right) \right) \right) \\
 &\quad \left(\tan^{-1}(x) - \tan^{-1}\left(\frac{x}{2}\right) + \left(\tan^{-1}\left(\frac{x}{2}\right) - \tan^{-1}\left(\frac{x}{3}\right) \right) \right) + \left(\tan^{-1}\left(\frac{x}{3}\right) - \tan^{-1}\left(\frac{x}{4}\right) \right) + \\
 &\quad \dots + \left(\tan^{-1}\left(\frac{x}{n}\right) - \tan^{-1}\left(\frac{x}{n+1}\right) \right) \\
 &= \tan^{-1}(x) - \tan^{-1}\left(\frac{x}{n+1}\right).
 \end{aligned}$$

$$\frac{dy}{dx} = \frac{1}{1+x^2} - \frac{1}{1+\left(\frac{x}{n+1}\right)^2} \cdot \left(\frac{1}{n+1}\right)$$

when $n \rightarrow \infty$

$$\frac{dy}{dx} = \frac{1}{1+x^2} - 0 = \frac{1}{1+x^2}.$$

Illustration :

$$(i) \quad y = \tan^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right) \quad (ii) \quad y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$$

$$(iii) \quad \text{If } f(x) = \sin^{-1}\left(\frac{2^{x+1}}{1+4^x}\right), \text{ find } f'(0).$$

Sol.

$$(i) \quad y = \tan^{-1}\left(\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right)$$

$$\text{Let } x = \cos 2\theta \Rightarrow \theta = \frac{1}{2} \cos^{-1}(x)$$

$$y = \tan^{-1}\left(\frac{\sqrt{1+\cos 2\theta}-\sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta}+\sqrt{1-\cos 2\theta}}\right) = \tan^{-1}\left(\frac{\sqrt{2}\cos\theta-\sqrt{2}\sin\theta}{\sqrt{2}\cos\theta+\sqrt{2}\sin\theta}\right) = \tan^{-1}\left(\frac{1-\tan\theta}{1+\tan\theta}\right)$$

$$y = \tan^{-1}\left(\tan\left(\frac{\pi}{4}-\theta\right)\right) = \frac{\pi}{4} - \theta$$

$$y = \frac{\pi}{4} - \frac{1}{2} \cos^{-1}x$$

$$\therefore \frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}}.$$

$$(ii) \quad y = \tan^{-1} \left(\frac{\sqrt{1+x^2} - 1}{x} \right)$$

Put $x = \tan \theta \therefore \theta = \tan^{-1} x$

$$y = \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) = \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) = \tan^{-1} \left(\frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right)$$

$$\Rightarrow y = \frac{\theta}{2} = \frac{1}{2} \tan^{-1} x \quad \Rightarrow \frac{dy}{dx} = \frac{1}{2(1+x^2)}$$

$$(iii) \quad f(x) = \sin^{-1} \left(\frac{2^{x+1}}{1+2^{2x}} \right)$$

Let $2^x = \tan \theta$

$$f(x) = \sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) = \sin^{-1} (\sin 2\theta)$$

$$f(x) = 2\theta = 2 \tan^{-1} (2^x) \Rightarrow f'(x) = \frac{1}{1+2^{2x}} (2^x \ln 2)$$

$$f'(0) = \frac{2}{1+1} (2^0 \ln 2) = \ln 2.$$

Derivative of $\sin^{-1} \left(\frac{2x}{1+x^2} \right)$; $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$; $\tan^{-1} \left(\frac{2x}{1-x^2} \right)$; $\sin^{-1}(3x - 4x^3)$;
 $\cos^{-1}(4x^3 - 3x)$:

$$(i) \quad y = f(x) = \sin^{-1} \left(\frac{2x}{1+x^2} \right) = \begin{cases} 2 \tan^{-1} x & |x| \leq 1 \\ \pi - 2 \tan^{-1} x & x > 1 \\ -(\pi + 2 \tan^{-1} x) & x < -1 \end{cases}$$

Highlights :

(a) Domain is $x \in \mathbb{R}$ &

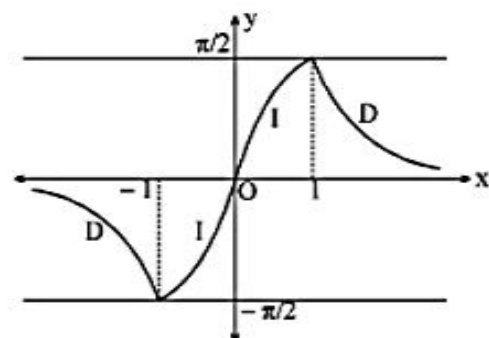
range is $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$

(b) f is continuous for all x but not diff. at $x = 1, -1$

$$(c) \quad \frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for } |x| < 1 \\ \text{nonexistent} & \text{for } |x| = 1 \\ -\frac{2}{1+x^2} & \text{for } |x| > 1 \end{cases}$$

(d) Increasing in $(-1, 1)$ & decreasing in $(-\infty, -1) \cup (1, \infty)$

Note: f is odd, aperiodic bound

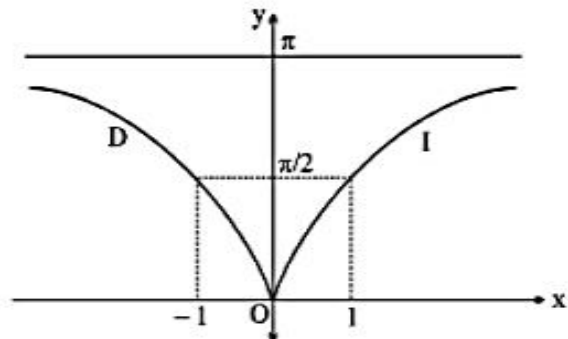


(ii) Consider $y = f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & \text{if } x \geq 0 \\ -2\tan^{-1}x & \text{if } x < 0 \end{cases}$

Highlights :

- (a) Domain is $x \in \mathbb{R}$ & range is $(0, \pi)$
 (b) Continuous for all x
 but not differentiable at $x=0$

(c) $\frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for } x > 0 \\ \text{nonexistent} & \text{for } x = 0 \\ -\frac{2}{1+x^2} & \text{for } x < 0 \end{cases}$



- (d) Increasing in $(0, \infty)$ & decreasing in $(-\infty, 0)$ Note: f is even, a periodic bound.

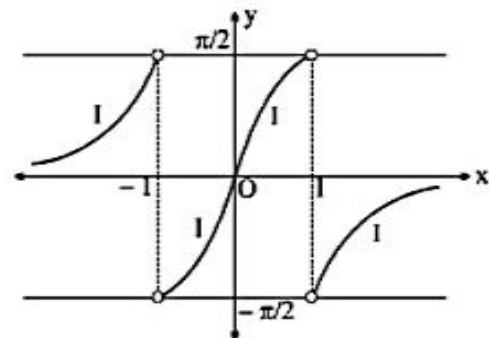
(iii) $y = f(x) = \tan^{-1} \frac{2x}{1-x^2} = \begin{cases} 2\tan^{-1}x & |x| < 1 \\ \pi + 2\tan^{-1}x & x < -1 \\ -(\pi - 2\tan^{-1}x) & x > 1 \end{cases}$

Highlights :

- (a) Domain is $\mathbb{R} - \{1, -1\}$ & range is $(-\frac{\pi}{2}, \frac{\pi}{2})$
 (b) f is neither continuous
 nor differentiable at $x = 1, -1$

(c) $\frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & |x| \neq 1 \\ \text{nonexistent} & |x| = 1 \end{cases}$

- (d) Increasing $\forall x$ in its domain
 (e) It is bound for all x



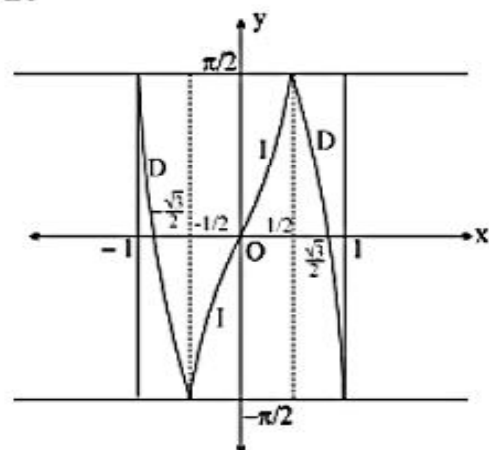
(iv) $y = f(x) = \sin^{-1}(3x - 4x^3) = \begin{cases} -(\pi + 3\sin^{-1}x) & \text{if } -1 \leq x \leq -\frac{1}{2} \\ 3\sin^{-1}x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \pi - 3\sin^{-1}x & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$

Highlights :

- (a) Domain is $x \in [-1, 1]$ & range is $[-\frac{\pi}{2}, \frac{\pi}{2}]$
 (b) Not derivable at $|x| = \frac{1}{2}$

(c) $\frac{dy}{dx} = \begin{cases} \frac{3}{\sqrt{1-x^2}} & \text{if } x \in (-\frac{1}{2}, \frac{1}{2}) \\ -\frac{3}{\sqrt{1-x^2}} & \text{if } x \in (-1, -\frac{1}{2}) \cup (\frac{1}{2}, 1) \end{cases}$

- (d) Continuous everywhere in its domain



$$(v) \quad y = f(x) = \cos^{-1}(4x^3 - 3x) = \begin{cases} 3\cos^{-1}x - 2\pi & \text{if } -1 \leq x \leq -\frac{1}{2} \\ 2\pi - 3\cos^{-1}x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ 3\cos^{-1}x & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

Highlights :

(a) Domain is $x \in [-1, 1]$ & range is $[0, \pi]$

(b) Continuous everywhere in its domain

but not derivable at $x = \frac{1}{2}, -\frac{1}{2}$

(c) Increasing in $\left(-\frac{1}{2}, \frac{1}{2}\right)$ &

Decreasing in $\left(\frac{1}{2}, 1\right] \cup \left[-1, -\frac{1}{2}\right)$

$$(d) \quad \frac{dy}{dx} = \begin{cases} \frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ -\frac{3}{\sqrt{1-x^2}} & \text{if } x \in \left(-1, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, 1\right) \end{cases}$$

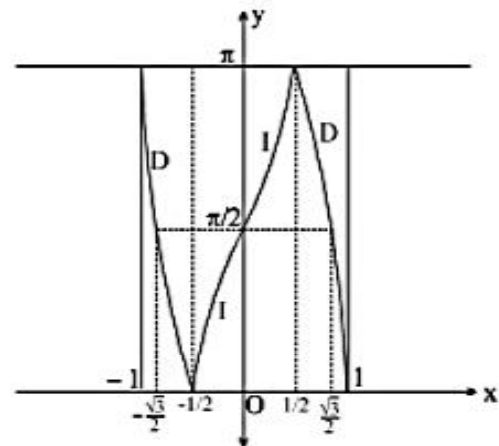


Illustration :

If $y = \sin^{-1} \left[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2} \right]$, and $0 < x < 1$, then find $\frac{dy}{dx}$

Sol. $y = \sin^{-1} \left[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2} \right]$, where $0 < x < 1$

$$= \sin^{-1} \left[x\sqrt{1-(\sqrt{x})^2} - \sqrt{x}\sqrt{1-x^2} \right] = \sin^{-1} x - \sin^{-1} \sqrt{x}$$

Differentiating w.r.t. x , we get

$$\frac{d}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{d}{dx} (\sqrt{x}) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x-x^2}}$$

Illustration :

Find $\frac{dy}{dx}$ for $y = \tan^{-1} \sqrt{\frac{a-x}{a+x}}$, where $-a < x < a$.

Sol. $y = \tan^{-1} \left\{ \sqrt{\frac{a-x}{a+x}} \right\}$, where $-a < x < a$

Substituting $x = a \cos \theta$, we have

$$y = \tan^{-1} \left\{ \sqrt{\frac{1-\cos\theta}{1+\cos\theta}} \right\} = \tan^{-1} \left\{ \sqrt{\tan^2 \frac{\theta}{2}} \right\} = \tan^{-1} \left| \tan \frac{\theta}{2} \right|.$$

Also for $-a < x < a$, $-1 < \cos \theta < 1$

$$\Rightarrow \theta \in (0, \pi) \Rightarrow \frac{\theta}{2} \in \left(0, \frac{\pi}{2}\right)$$

$$\therefore y = \tan^{-1} \left| \tan \frac{\theta}{2} \right| = \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \frac{\theta}{2} = \frac{1}{2} \cos^{-1} \left(\frac{x}{a} \right)$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{2} \cdot \frac{1}{\sqrt{1 - \frac{x^2}{a^2}}} \cdot \frac{d}{dx} \left(\frac{x}{a} \right) = -\frac{1}{2\sqrt{a^2 - x^2}}$$

Practice Problem

Q.1 The function $f(x) = e^x + x$, being differentiable and one to one, has a differentiable inverse $f^{-1}(x)$. The value of $\frac{d}{dx}(f^{-1})$ at the point $f(\log 2)$ is

- (A) $\frac{1}{\ln 2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) None of these

Q.2 Let $g(x)$ be the inverse of an invertible function $f(x)$ which is differentiable at $x = c$, then $g'(f(c))$ equals

- (A) $f'(c)$ (B) $\frac{1}{f'(c)}$ (C) $f(c)$ (D) None of these

Q.3 If $f(x) = x + \tan x$ and f is inverse of g , then $g'(x)$ equals

- (A) $\frac{1}{1+[g(x)-x]^2}$ (B) $\frac{1}{2-[g(x)-x]^2}$ (C) $\frac{1}{2+[g(x)-x]^2}$ (D) None of these

Q.4 Find $\frac{dy}{dx}$ of the following functions :

(i) $y = \sec^{-1} \left(\frac{\sqrt{x}+1}{\sqrt{x}-1} \right) + \sin^{-1} \left(\frac{\sqrt{x}-1}{\sqrt{x}+1} \right)$ (ii) $y = \tan^{-1} \frac{4x}{1+5x^2} + \tan^{-1} \frac{2+3x}{3-2x}$

(iii) $y = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$, where $x \neq 0$ (iv) $y = \tan^{-1} \frac{3a^2x - x^3}{a(a^2 - 3x^2)}$

(v) $y = \sin^{-1} \left(\frac{5x+12\sqrt{(1-x)^2}}{13} \right)$ (vi) $y = \tan^{-1} \left(\frac{x}{1+\sqrt{1-x^2}} \right)$

(vii) $y = \tan^{-1} \left(\frac{a \cos x - b \sin x}{b \cos x + a \sin x} \right)$, where $-\frac{\pi}{2} < x < \frac{\pi}{2}$ and $\frac{a}{b} \tan x > -1$.

$$(viii) \quad y = \tan^{-1} \left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right), \text{ where } -1 < x < 1, x \neq 0.$$

$$(ix) \quad y = \sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right) + \cos^{-1} \left(\frac{1}{\sqrt{1+x^2}} \right), 0 < x < \infty.$$

$$(x) \quad y = (\sin x)^x + \sin^{-1} \sqrt{x}$$

Q.5 Find the derivative of the following functions with respect to x :

$$(i) \quad y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$(ii) \quad y = \tan^{-1} \left(\frac{3x-x^3}{1-3x^2} \right), \frac{-1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$$

$$(iii) \quad y = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right), 0 < x < 1$$

$$(iv) \quad y = \sin^{-1} \left(\frac{1-x^2}{1+x^2} \right), 0 < x < 1$$

$$(v) \quad y = \cos^{-1} \left(\frac{2x}{1+x^2} \right), -1 < x < 1$$

$$(vi) \quad y = \sin^{-1} \left(2x\sqrt{1-x^2} \right), -\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$$

$$(vii) \quad y = \sec^{-1} \left(\frac{1}{2x^2-1} \right), 0 < x < \frac{1}{\sqrt{2}}$$

Q.6 Find the derivative of the following functions with respect to x :

$$(i) \quad \cos^{-1}(\sin x)$$

$$(ii) \quad \tan^{-1} \left(\frac{\sin x}{1+\cos x} \right)$$

$$(iii) \quad \sin^{-1} \left(\frac{2^{x+1}}{1+4^x} \right)$$

Answer key

Q.1 B

Q.2 B

Q.3 C

Q.4 (i) 0 (ii) $\frac{1}{1+25x^2}$ (iii) $\frac{1}{2(1+x^2)}$ (iv) $\frac{3a}{(a^2+x^2)}$

(v) $\frac{1}{\sqrt{1-x^2}}$ (vi) $\frac{1}{2\sqrt{1-x^2}}$ (vii) -1 (viii) $-\frac{x}{\sqrt{1-x^4}}$

(ix) $\frac{2}{(1+x^2)}$ (x) $(\sin x)^x (x \cot x + \ln \sin x) + \frac{1}{2\sqrt{x(1-x)}}$

T-9 PARAMETRIC DIFFERENTIATION :

In some situation curves are represented by the equations e.g. $x = \sin t$ & $y = \cos t$

If $x = f(t)$ and $y = g(t)$ then

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{g'(t)}{f'(t)}$$

Illustration :

Find $\frac{dy}{dx}$ if $x = a(\theta - \sin \theta)$ and $y = a(1 - \cos \theta)$.

Sol. We have,

$$x = a(\theta - \sin \theta) \text{ and } y = a(1 - \cos \theta)$$

$$\Rightarrow \frac{dy}{dx} = a(1 - \cos \theta) \text{ and } \frac{dy}{d\theta} = a \sin \theta$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{a \sin \theta}{a(1 - \cos \theta)} = \frac{2 \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right)}{2 \sin^2\left(\frac{\theta}{2}\right)} = \cot \frac{\theta}{2}.$$

Illustration :

If $x = a \sec^3 \theta$ and $y = a \tan^3 \theta$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$.

Sol. We have $x = a \sec^3 \theta$ and $y = a \tan^3 \theta$

$$\frac{dx}{d\theta} = 3a \sec^2 \theta \frac{d}{d\theta} (\tan \theta) = 3a \tan^2 \theta \sec^2 \theta$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{3a \tan^2 \theta \sec^2 \theta}{3a \tan^3 \theta \sec^2 \theta} = \frac{\tan \theta}{\sec \theta} = \sin \theta$$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{\theta=\frac{\pi}{3}} = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}.$$

Illustration :

Find $\frac{dy}{dx}$ if

$$(i) \quad x = a(\cos t + t \sin t) \text{ and } y = a(\sin t - t \cos t) \quad (ii) \quad x = \frac{3at}{1+t^3}; y = \frac{3at^2}{1+t^3}$$

$$(iii) \quad x = a \sec^2 \theta; y = a \tan^3 \theta.$$

$$(iv) \quad x = a\sqrt{\cos 2t} \cos t \text{ and } y = a\sqrt{\cos 2t} \sin t \text{ then, find } \frac{dy}{dx} \Big|_{t=\pi/6}.$$

$$(v) \quad \text{If } x = \sec \theta - \cos \theta \text{ \& } y = \sec^n \theta - \cos^n \theta, \text{ then show that } (x^2 + 4) \left(\frac{dy}{dx}\right)^2 = n^2(y^2 + 4).$$

Sol.

(i) $x = a (\cos t + t \sin t)$ and $y = a (\sin t - t \cos t)$

$$\frac{dx}{dt} = a (-\sin t + \sin t + t \cos t) = at \cos t$$

$$\frac{dy}{dt} = a (\cos t - \sin t + t \sin t) = at \sin t$$

$$\Rightarrow \frac{dy}{dx} = \tan t$$

(ii) $x = \frac{3at}{1+t^3}; y = \frac{3at^2}{1+t^3}$

$$\frac{dx}{dt} = \frac{3a(1+t^3) \cdot 1 - t(3t^2)}{(1+t^3)^2} = \frac{3a(1-2t^3)}{(1+t^3)^2}$$

$$\frac{dy}{dt} = \frac{3a(1+t^3) \cdot 2t - t^2(3t^2)}{(1+t^3)^2} = \frac{3a(2t-t^4)}{(1+t^3)^2}$$

$$\frac{dy}{dx} = \frac{(2-t^3)t}{(1-2t^3)}$$

(iii) $x = a \sec^2 \theta; y = a \tan^3 \theta$

$$\frac{dx}{d\theta} = 2a \sec \theta (\sec \theta \tan \theta)$$

$$\frac{dy}{d\theta} = 3a \tan^2 \theta \sec^2 \theta$$

$$\frac{dy}{dx} = \frac{3 \tan \theta}{2}$$

(iv) $x = a \sqrt{\cos 2t} \cos t$ and $y = a \sqrt{\cos 2t} \sin t$

$$\frac{dx}{dt} = a \left(\sqrt{\cos 2t} (-\sin t) + \frac{\cos t (-2 \sin 2t)}{2 \sqrt{\cos 2t}} \right)$$

$$\frac{dx}{dt} = a \left(\frac{\sin t \cos 2t + \cos t \sin 2t}{\sqrt{\cos 2t}} \right)$$

$$\frac{dx}{dt} = -a \frac{\sin 3t}{\sqrt{\cos 2t}} \Rightarrow \left. \frac{dx}{dt} \right|_{\pi/6} = -\sqrt{2} a$$

$$\frac{dy}{dt} = a \left(\sqrt{\cos 2t} \cos t + \frac{\sin t}{2 \sqrt{\cos 2t}} (-2 \sin 2t) \right) = a \left(\frac{\cos 3t}{\sqrt{\cos 2t}} \right) \Rightarrow \left. \frac{dy}{dt} \right|_{\pi/6} = 0$$

$$\left. \frac{dy}{dx} \right|_{\pi/6} = 0.$$

(v) $x = \sec \theta - \cos \theta ; y = \sec^n \theta - \cos^n \theta$

$$(x^2 + 4) \left(\frac{dy}{dx} \right)^2 = n^2 (y^2 + 4)$$

$$\frac{dx}{d\theta} = \sec \theta \tan \theta + \sin \theta = \sin \theta (\sec^2 \theta + 1)$$

$$\frac{dy}{d\theta} = n \sec^{n-1} \theta \sec \theta \tan \theta + n \cos^{n-1} \theta \sin \theta = n \sec^n \theta \tan \theta + n \cos^n \theta \tan \theta$$

$$\frac{dy}{d\theta} = n \tan \theta (\sec^n \theta + \cos^n \theta)$$

$$\frac{dy}{dx} = \frac{n \tan \theta (\sec^n \theta + \cos^n \theta)}{\sin \theta (\sec^2 \theta + 1)} = \frac{n (\sec^n \theta + \cos^n \theta)}{\cos \theta (\sec^2 \theta + 1)} = \frac{n (\sec^n \theta + \cos^n \theta)}{(\sec \theta + \cos \theta)}$$

$$\left(\frac{dy}{dx} \right)^2 = \frac{n^2 (\sec^{2n} \theta + \cos^{2n} \theta + 2)}{(\sec^2 \theta + \cos^2 \theta + 2)} = \frac{n^2 (y^2 + 4)}{(x^2 + 4)}$$

Illustration :

For the curve represented parametrically indicate the relation between the parameter t and the angle α between the tangent to the given curve and the x -axis.

$$(i) \quad \begin{cases} x = \cos t + t \sin t - \frac{t^2}{2} \cos t \\ y = \sin t - t \cos t - \frac{t^2}{2} \sin t \end{cases} ; \quad (ii) \quad x = a \cos^3 t, y = a \sin^3 t$$

Sol.

(i) $x = \cos t + t \sin t - \frac{t^2}{2} \cos t$

$$\frac{dx}{dt} = -\sin t + t \cos t + \sin t - \left(\frac{t^2}{2} \sin t + t \cos t \right) = \frac{t^2 \sin t}{2}$$

$$y = \sin t - t \cos t - \frac{t^2}{2} \sin t$$

$$\frac{dy}{dt} = \cos t - (\cos t - t \sin t) - \frac{1}{2} (2t \sin t + t^2 \cos t) = \frac{-t^2 \cos t}{2}$$

$$\therefore \frac{dy}{dx} = \frac{(dy/dt)}{(dx/dt)} = -\cot t = \tan \alpha \Rightarrow \tan \left(\frac{\pi}{2} - t \right) = \tan (-\alpha)$$

$$\Rightarrow \frac{\pi}{2} - t = -\alpha \Rightarrow t = \frac{\pi}{2} + \alpha$$

$$\begin{aligned}
 \text{(ii)} \quad x &= a \cos^3 t \Rightarrow \frac{dx}{dt} = -3a \cos^2 t \sin t \\
 y &= a \sin^3 t \Rightarrow \frac{dy}{dt} = 3a \sin^2 t \cos t \\
 \therefore \frac{dy}{dx} &= \frac{(dy/dt)}{(dx/dt)} = \frac{3a \sin^2 t \cos t}{-3a \cos^2 t \sin t} = -\tan t = \tan \alpha \\
 \Rightarrow \tan(\pi - t) &= \tan \alpha \Rightarrow \pi - t = \alpha \Rightarrow t = \pi - \alpha.
 \end{aligned}$$

T-10 DIFFERENTIATION OF ONE FUNCTION W.R.T. OTHER FUNCTION :

If $y = f(x)$ and $z = g(x)$ then derivative of $f(x)$ w.r.t. $g(x)$ is given by

$$\begin{aligned}
 \frac{dy}{dz} &= \frac{dy}{dx} \cdot \frac{dx}{dz} = \frac{f'(x)}{g'(x)} \\
 \therefore \text{Differential coefficient of } f(x) \text{ w.r.t. } g(x) &= \frac{\text{derivative of } f(x) \text{ w.r.t. } x}{\text{derivative of } g(x) \text{ w.r.t. } x} = \frac{f'(x)}{g'(x)}
 \end{aligned}$$

Illustration :

Differentiate $\log \sin x$ w.r.t. $\sqrt{\cos x}$.

Sol. Let $u = \log \sin x$ and $v = \sqrt{\cos x}$

$$\text{Then, } \frac{du}{dx} = \cot x \text{ and } \frac{dv}{dx} = -\frac{\sin x}{2\sqrt{\cos x}}$$

$$\Rightarrow \frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{\cot x}{-\frac{\sin x}{2\sqrt{\cos x}}} = -2\sqrt{\cos x} \cot x \operatorname{cosec} x.$$

Illustration :

Differentiate $\tan^{-1} \left(\frac{\sqrt{1+x^2} - 1}{x} \right)$ w.r.t. $\tan^{-1} x$,

Sol. Let $u = \tan^{-1} \left(\frac{\sqrt{1+x^2} - 1}{x} \right)$ and $v = \tan^{-1} x$.

Putting $x = \tan \theta$,

$$\begin{aligned}
 \text{we get } u &= \tan^{-1} \left(\frac{\sqrt{1+x^2} - 1}{x} \right) \\
 &= \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) = \tan^{-1} \left(\frac{1 - \cos \theta}{\sin \theta} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right) = \frac{1}{2} \theta = \frac{1}{2} \tan^{-1} x.
 \end{aligned}$$

Thus, we have $u = \frac{1}{2} \tan^{-1} x$ and $v = \tan^{-1} x$.

$$\Rightarrow \frac{du}{dx} = \frac{1}{2} \times \frac{1}{1+x^2} \text{ and } \frac{dv}{dx} = \frac{1}{1+x^2}$$

$$\Rightarrow \frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{1}{2(1+x^2)} (1+x^2) = \frac{1}{2}.$$

Illustration :

Find derivative of $(\ln x)^{\tan x}$ w.r.t. x^x .

Sol. $u = (\ln x)^{\tan x}$; $v = x^x$
 $\ln(u) = \tan x \ln(\ln x)$

$$\Rightarrow \frac{1}{u} \left(\frac{du}{dx} \right) = (\sec^2 x) \ln(\ln x) + \tan x \left(\frac{1}{\ln x} \cdot \frac{1}{x} \right)$$

$$\frac{du}{dx} = \frac{u \left(x \ln x \ln(\ln x) \sec^2 x + \tan x \right)}{(x \ln x)}$$

$$\ln v = x \ln x$$

$$\Rightarrow \frac{1}{v} \left(\frac{dv}{dx} \right) = (\ln x + 1)$$

$$\frac{du}{dx} = \frac{(\ln x)^{\tan x}}{x^x} \left(\frac{x \ln x \ln(\ln x) \sec^2 x + \tan x}{(x \ln x)(\ln x + 1)} \right).$$

Illustration :

Derivative of $\cos^{-1}(2x^2 - 1)$ w.r.t. $\sqrt{1-x^2}$ when $x = \frac{1}{2}$.

Sol. $u = \cos^{-1}(2x^2 - 1)$; $v = \sqrt{1-x^2}$

$$\text{put } x = \cos \theta \text{ as } x = \frac{1}{2} \text{ so } \theta = \frac{\pi}{3}$$

$$u = \cos^{-1}(\cos 2\theta) = 2\theta \quad \text{put } v = \sin \theta$$

$$\frac{du}{dv} = \frac{du/d\theta}{dv/d\theta} = \frac{2}{\cos \theta} = 4.$$

Illustration :

Define derivative of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r.t. $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ $\forall x \in R$.

Sol. Let $u = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ and $v = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

Let $x = \tan \theta$

$$u = \sin^{-1}(\sin 2\theta) \quad \text{and} \quad v = \cos^{-1}(\cos 2\theta)$$

$$u = \begin{cases} 2\theta, & \frac{-\pi}{2} \leq 2\theta \leq \frac{\pi}{2} \Rightarrow \frac{-\pi}{4} \leq \theta \leq \frac{\pi}{4} \Rightarrow -1 \leq x \leq 1 \\ (-\pi - 2\theta), & \frac{\pi}{2} \leq 2\theta \leq \frac{3\pi}{2} \Rightarrow \frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4} \Rightarrow 1 \leq x \leq \infty \\ (\pi - 2\theta), & \frac{3\pi}{2} \leq 2\theta \leq \frac{5\pi}{2} \Rightarrow \frac{3\pi}{4} \leq \theta \leq \frac{5\pi}{4} \Rightarrow -\infty \leq x \leq -1 \end{cases}$$

$$v = \cos^{-1}(\cos 2\theta)$$

$$v = \begin{cases} 2\theta, & 0 < 2\theta \leq \pi \Rightarrow 0 < \theta \leq \frac{\pi}{2} \Rightarrow 0 < x < \infty \\ -2\theta, & -\pi < 2\theta \leq 0 \Rightarrow \frac{-\pi}{2} \leq \theta \leq 0 \Rightarrow -\infty < x < 0 \end{cases}$$

$$\frac{du}{d\theta} = \begin{cases} 2, & -1 \leq x \leq 1 \\ -2, & 1 \leq x < \infty \\ -2, & -\infty < x \leq -1 \end{cases}$$

$$\frac{dv}{d\theta} = \begin{cases} 2, & 0 < x < \infty \\ -2, & -\infty < x < 0 \end{cases}$$

$$\frac{du}{dv} = \begin{cases} \left(\frac{-2}{-2}\right) = 1, & -\infty < x < -1 \\ \left(\frac{2}{-2}\right) = -1, & -1 < x < 0 \\ \left(\frac{2}{2}\right) = 1, & 0 < x < 1 \\ \left(\frac{-2}{2}\right) = -1, & 1 < x < \infty \end{cases}$$

$$\frac{du}{dv} = \begin{cases} 1 \quad \forall x \in (-\infty, -1) \cup (0, 1) \\ -1 \quad \forall x \in (-1, 0) \cup (1, \infty) \\ \text{not exists at } x = -1, 0, 1 \end{cases}$$

Illustration :

- (i) Differential coefficient of $e^{\sin^{-1} x}$ w.r.t. $e^{-\cos^{-1} x}$ is independent of x . (T or F)
 (ii) Find the derivative of $f(\tan x)$ w.r.t. $g(\sec x)$ at $x = \frac{\pi}{4}$, where $f'(1) = 2$ and $g'(\sqrt{2}) = 4$.

Sol.

- (i) Let $u = e^{\sin^{-1} x}$ and $v = e^{-\cos^{-1} x}$

$$u = e^{\frac{\pi}{2} - \cos^{-1} x} \quad \left[\because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \right]$$

$$u = e^{\frac{\pi}{2}} e^{-\cos^{-1} x} = e^{\frac{\pi}{2}} v$$

$$\text{So, } \frac{du}{dv} = e^{\frac{\pi}{2}}. \text{ (True)}$$

- (ii) Let $u = f(\tan x)$ and $v = g(\sec x)$

$$\Rightarrow \frac{du}{dx} = f'(\tan x) \sec^2 x \quad \text{and} \quad \frac{dv}{dx} = g'(\sec x) \sec x \tan x$$

$$\Rightarrow \frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{f'(\tan x) \sec^2 x}{g'(\sec x) \sec x \tan x}$$

$$\Rightarrow \left[\frac{du}{dv} \right]_{x=\frac{\pi}{4}} = \frac{f'\left(\tan \frac{\pi}{4}\right)}{g'\left(\sec \frac{\pi}{4}\right) \sin \frac{\pi}{4}} = \frac{f'(1) \sqrt{2}}{g'(\sqrt{2})} = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}}.$$

General Note :

Concavity in each case is decided by the sign of 2nd derivative as :

$$\frac{d^2y}{dx^2} > 0 \Rightarrow \text{Concave upwards} ; \quad \frac{d^2y}{dx^2} < 0 \Rightarrow \text{Concave downwards}$$

D = Decreasing; I = Increasing

T-11 SUCCESSIVE DIFFERENTIATION :

$y = f(x)$; the popular symbols used to denote the derivatives

are $\frac{dy}{dx} = Dy = f'(x) = y_1 = y'$. Higher order derivatives are

denoted as $\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2} = D^2y = f''(x) = y_2$ or y'' etc.

Note : A homogeneous equation of degree n represents ' n ' straight lines passing through the origin, hence

$$\frac{dy}{dx} = \frac{y}{x} \quad \text{and} \quad \frac{d^2y}{dx^2} = 0$$

$$\text{e.g. If } x^3 + 3x^2y - 6xy^2 + 2y^3 = 0, \text{ then } \left[\frac{d^2y}{dx^2} \right]_{(1,1)} = 0$$

Illustration :

If $y = \sin(\sin x)$ then prove that $y_2 + (\tan x)y_1 + y \cos^2 x = 0$.

Sol. $y = \sin(\sin x)$

$$y_1 = \cos(\sin x) \cdot \cos x$$

$$y_2 = -\cos(\sin x) \sin x - \sin(\sin x) \cos^2 x$$

$$\therefore y_2 + (\tan x) y_1 + y \cos^2 x$$

$$= -\cos(\sin x) \cdot \sin x - \sin(\sin x) \cos^2 x + \tan x (\cos(\sin x) \cdot \cos x) + \sin(\sin x) \cos^2 x$$

$$= 0.$$

Illustration :

(i) If $y = a \cos(\ln x) + b \sin(\ln x)$ then prove that $x^2 y_3 + 3xy_2 + 2y_1 = 0$

(ii) If $y = e^{a \cos^{-1} x}$, then prove that $(1 - x^2)y_2 - xy_1 - a^2 y = 0$.

(iii) If $y^{\frac{1}{m}} + y^{-\frac{1}{m}} = 2x$ then prove that $(x^2 - 1)y_3 + 3xy_2 + (1 - m^2)y_1 = 0$

Sol. (i) $y = a \cos(\ln x) + b \sin(\ln x)$

$$y_1 = \frac{-a \sin(\ln x)}{x} + \frac{b \cos(\ln x)}{x}$$

$$\Rightarrow x y_1 = -a \sin(\ln x) + b \cos(\ln x) \Rightarrow x y_2 + y_1 = \frac{-a \cos(\ln x)}{x} - \frac{b \sin(\ln x)}{x}$$

$$\Rightarrow x^2 y_2 + x y_1 = -a \cos(\ln x) - b \sin(\ln x) = -y \Rightarrow x^2 y_2 + x y_1 + y = 0$$

$$\Rightarrow (x^2 y_3 + 2xy_2) + (x y_2 + y_1) + y_1 = 0$$

$$\Rightarrow x^2 y_3 + 3xy_2 + 2y_1 = 0.$$

(ii) $y = e^{a \cos^{-1} x}$

$$\frac{dy}{dx} = e^{a \cos^{-1} x} \left(\frac{-a}{\sqrt{1-x^2}} \right) = y_1$$

$$\Rightarrow y_1 = \frac{-ay}{\sqrt{1-x^2}} \Rightarrow (1-x^2)(y_1)^2 = a^2 y^2$$

now differentiating w.r.t. to x

$$(1-x^2) \cdot 2y_1 y_2 - 2x (y_1)^2 = a^2 (2yy_1)$$

$$\Rightarrow (1-x^2) y_2 - x y_1 = a^2 y.$$

$$\begin{aligned}
 \text{(iii)} \quad & \frac{1}{y^m} + \frac{-1}{y^{m-1}} = 2x \\
 \Rightarrow & y^{2/m} - 2x y^{1/m} + 1 = 0 \Rightarrow y^{1/m} = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = \left(x \pm \sqrt{x^2 - 1} \right) \\
 \Rightarrow & y = \left(x \pm \sqrt{x^2 - 1} \right)^m \Rightarrow y_1 = m \left(x \pm \sqrt{x^2 - 1} \right)^{m-1} \left(1 \pm \frac{x}{\sqrt{x^2 - 1}} \right) \\
 \Rightarrow & y_1 = \frac{m}{\sqrt{x^2 - 1}} \left(x \pm \sqrt{x^2 - 1} \right)^{m-1} \left(\sqrt{x^2 - 1} \pm x \right) \\
 \Rightarrow & y_1 = \pm \frac{my}{\sqrt{x^2 - 1}} \\
 \Rightarrow & (x^2 - 1) (y_1)^2 = m^2 y^2 \Rightarrow (x^2 - 1) 2y y_2 + (y_1)^2 2x = m^2 y y_1 \\
 \Rightarrow & (x^2 - 1) y_2 + x y_1 = m^2 y \\
 \Rightarrow & (x^2 - 1) y_3 + 2x y_2 + x y_2 + y_1 = m^2 y_1 \Rightarrow (x^2 - 1) y_3 + 3x y_2 + (1 - m^2) y_1 = 0.
 \end{aligned}$$

Illustration :

If $(x-a)^2 + (y-b)^2 = c^2$ ($c > 0$) then $\frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\frac{d^2y}{dx^2}}$ equals

- (A) c (B) c^2 (C) c^3 (D) c^4

Sol. $(x-a)^2 + (y-b)^2 = c^2$ ($c > 0$)(1)

Now differentiating w.r.t. to x

$$2(x-a) + 2(y-b) y_1 = 0$$

$$\Rightarrow (x-a) + (y-b) y_1 = 0 \quad \text{.....(2)}$$

$$\Rightarrow 1 + (y-b) y_2 + (y_1)^2 = 0 \quad \text{.....(3)}$$

$$1 + (y_1)^2 = 1 + \frac{(x-a)^2}{(y-b)^2} \quad \text{from (2)}$$

$$1 + (y_1)^2 = \frac{c^2}{(y-b)^2}$$

$$\begin{aligned}
 \frac{(1 + (y_1)^2)^{3/2}}{y_2} &= \frac{\left(\frac{c^2}{(y-b)^2} \right)^{3/2}}{-\frac{(1 + (y_1)^2)}{(y-b)}} \quad \text{from (3)} \\
 &= \frac{\left| \frac{c^3}{(y-b)^3} \right|}{-c^2/(y-b)^3} = \pm c.
 \end{aligned}$$

Illustration :

Use the substitution $x = \tan \theta$ to show that the equation,

$$\frac{d^2 y}{dx^2} + \frac{2x}{1+x^2} \frac{dy}{dx} + \frac{y}{(1+x^2)^2} = 0 \text{ changes to } \frac{d^2 y}{d\theta^2} + y = 0.$$

Sol. $\frac{d^2 y}{dx^2} + \frac{2x}{1+x^2} \left(\frac{dy}{dx} \right) + \frac{y}{(1+x^2)^2} = 0 \quad \dots\dots(1)$

$$x = \tan \theta \Rightarrow dx = \sec^2 \theta d\theta \Rightarrow \frac{d\theta}{dx} = \frac{1}{\sec^2 \theta} = \cos^2 \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \left(\frac{d\theta}{dx} \right) = \frac{1}{\sec^2 \theta} \left(\frac{dy}{d\theta} \right) = \cos^2 \theta \left(\frac{dy}{d\theta} \right)$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{\sec^2 \theta} \cdot \frac{dy}{d\theta} \right) = \frac{d}{d\theta} \left(\cos^2 \theta \frac{dy}{d\theta} \right) \cdot \frac{d\theta}{dx} = \left(\cos^2 \theta \frac{d^2 y}{d\theta^2} - 2 \cos \theta \sin \theta \frac{dy}{d\theta} \right) \cos^2 \theta$$

From equation (1) putting $x = \tan \theta$

$$\cos^2 \theta \left(\cos^2 \theta \frac{d^2 y}{d\theta^2} - 2 \cos \theta \sin \theta \frac{dy}{d\theta} \right) + \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) \left(\cos^2 \theta \frac{dy}{d\theta} \right) + \left(\frac{y}{(1 + \tan^2 \theta)^2} \right) = 0$$

$$\Rightarrow \left(\frac{d^2 y}{d\theta^2} - 2 \tan \theta \frac{dy}{d\theta} \right) + \left(2 \tan \theta \frac{dy}{d\theta} \right) + y = 0 \Rightarrow \frac{d^2 y}{d\theta^2} + y = 0.$$

Illustration :

Starting with $\frac{dx}{dy} = \frac{1}{dy/dx}$. Prove that $\frac{d^2 x}{dy^2} = - \frac{d^2 y/dx^2}{(dy/dx)^3}$ and deduce that for the parabola

$$y^2 = 4ax, \quad \frac{d^2 y}{dx^2} \cdot \frac{d^2 x}{dy^2} = - \frac{2a}{y^3}$$

Sol. $\frac{dx}{dy} = \frac{1}{\left(\frac{dy}{dx} \right)}$

$$\frac{d^2 x}{dy^2} = \frac{d}{dy} \left(\frac{1}{\left(\frac{dy}{dx} \right)} \right) = \frac{1}{dx} \left(\left(\frac{dy}{dx} \right)^{-1} \right) \cdot \frac{dx}{dy} = -1 \left(\frac{dy}{dx} \right)^{-2} \left(\frac{d^2 y}{dx^2} \right) \cdot \frac{1}{\left(\frac{dy}{dx} \right)} = \frac{- \left(\frac{d^2 y}{dx^2} \right)}{\left(\frac{dy}{dx} \right)^3}$$

Given $y^2 = 4ax$

$$\Rightarrow 2y \frac{dy}{dx} = 4a \Rightarrow y \frac{dy}{dx} = 2a \Rightarrow \frac{dy}{dx} = \frac{2a}{y} \Rightarrow \frac{d^2y}{dx^2} = \frac{-2a}{y^2} \left(\frac{dy}{dx} \right) = \frac{-4a^2}{y^3}$$

$$\left(\frac{d^2y}{dx^2} \right) \left(\frac{d^2x}{dy^2} \right) = \left(\frac{d^2y}{dx^2} \right) \left(\frac{-\left(\frac{d^2y}{dx^2} \right)}{\left(\frac{dy}{dx} \right)^3} \right) = \frac{-\left(\frac{4a^2}{y^3} \right)^2}{\left(\frac{2a}{y} \right)^3} = \frac{-2a}{y^3}.$$

Illustration :

If $y = \left(\frac{1}{x} \right)^x$ then prove that $y_2(1) = 0$ i.e. $\frac{d^2y}{dx^2} = 0$

Sol. $y = x^{-x} \Rightarrow \ln y = -x \ln x$

$$\frac{1}{y} \left(\frac{dy}{dx} \right) = -(1 + \ln x) \Rightarrow \frac{dy}{dx} = -y(1 + \ln x) \quad \text{so } \frac{dy}{dx} \Big|_{x=1} = -y$$

$$\frac{d^2y}{dx^2} = - \left(\frac{dy}{dx} \right) (1 + \ln x) - \frac{y}{x}$$

$$\frac{d^2y}{dx^2} \Big|_{\text{at } x=1} = - \frac{dy}{dx} \Big|_{x=1} - y = y - y = 0.$$

Illustration :

If $e^{x+y} = y^2$ then prove that $y'' = \frac{2y}{(2-y)^3}$.

Sol. $e^{x+y} = y^2 \Rightarrow x + y = 2 \ln y$

$$1 + \frac{dy}{dx} = \frac{2}{y} \left(\frac{dy}{dx} \right)$$

$$1 = \left(\frac{2-y}{y} \right) \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{y}{2-y}$$

$$\frac{d^2y}{dx^2} = \frac{(2-y) \frac{dy}{dx} - y \left(\frac{-dy}{dx} \right)}{(2-y)^2} = \frac{2}{(2-y)^2} \left(\frac{dy}{dx} \right) = \frac{2y}{(2-y)^3}.$$

Illustration :

- (i) Use induction to prove that $D(x^n) = n x^{n-1} \quad \forall n \in \mathbb{N}$
 (ii) If $y = e^{\tan^{-1} x}$ then prove that $(1+x^2)y_{n+1} + (2nx-1)y_n + n(n-1)y_{n-1} = 0$.
 (iii) If $y = \sin(m \sin^{-1} x)$ then show that $(1-x^2)y_{n+2} = (2n+1)xy_{n+1} + (n^2-m^2)y_n$.

Sol.

- (i) $D(x^n) = n x^{n-1}$
 $D(x^{n+1}) = D(x^n \cdot x) = D(x^n) x + x^n D(x)$
 $= nx^{n-1} x + x^n = (n+1)x^n$
 (ii) $y = e^{\tan^{-1} x}$ then $(1+x^2)y_{n+1} + (2nx-1)y_n + n(n-1)y_{n-1} = 0$

$$\frac{dy}{dx} = \frac{e^{\tan^{-1} x}}{(1+x^2)} \Rightarrow (1+x^2)y_1 = e^{\tan^{-1} x} = y$$

Differentiating again

$$(1+x^2)y_2 + 2xy_1 = y_1$$

so at $n=1$; $(1+x^2)y_2 + (2x-1)y_1 + 0 = 0$.

Let this equation is true for n ; then

$$(1+x^2)y_{n+1} + (2nx-1)y_n + n(n-1)y_{n-1} = 0$$

Now again differentiating w.r.t. to x .

$$(1+x^2)y_{n+2} + 2xy_{n+1} + (2nx-1)y_{n+1} + 2ny_n + n(n-1)y_n = 0$$

$$(1+x^2)y_{n+2} + [2x(n+1)-1]y_{n+1} + n(n-1)y_n = 0$$

so the equation is valid for $(n+1)$ also

So from M.I. $(1+x^2)y_{n+1} + (2nx-1)y_n + n(n+1)y_{n-1} = 0$ is valid for all n .

- (iii) $y = \sin(m \sin^{-1} x)$ then $(1-x^2)y_{n+2} = (2n+1)xy_{n+1} + (n^2-m^2)y_n$
 Now, differentiating w.r.t. to x

$$\frac{dy}{dx} = \cos(m \sin^{-1} x) \frac{m}{\sqrt{1-x^2}}$$

$$y_1 = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}} \frac{m}{\sqrt{1-x^2}} \quad \text{Now, squaring we get}$$

$$y_1^2 = m^2 \frac{1-y^2}{1-x^2} \Rightarrow (1-x^2)y_1^2 = m^2(1-y^2)$$

$$(1-x^2)2y_1y_2 - 2xy_1^2 = -m^2 2yy_1 \Rightarrow (1-x^2)y_2 - xy_1 + m^2y = 0$$

$$\text{So for } n=0; (1-x^2)y_2 - xy_1 + m^2y = 0$$

Let the equation $(1-x^2)y_{n+2} = (2n+1)xy_{n+1} + (n^2-m^2)y_n$ is true for any n then again differentiating

$$(1-x^2)y_{n+3} - 2xy_{n+2} = (2n+1)xy_{n+2} + (2n+1)y_{n+1} + (n^2-m^2)y_{n+1}$$

$$\Rightarrow (1-x^2)y_{n+3} - 2xy_{n+2} - (2n+1)xy_{n+2} = [(n+1)^2 - m^2]y_{n+1}$$

$$\Rightarrow (1-x^2)y_{n+3} = [2(n+1)+1]xy_{n+2} + [(n+1)^2 - m^2]y_{n+1}$$

Practice Problem

Q.1 Find $\frac{dy}{dx}$, if

(i) $x = a \cos \theta, y = a \sin \theta$

(ii) $x = at^2, y = 2at$

(iii) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

(iv) $x = 2at^2, y = at^4$

(v) $x = \frac{\sin^3 t}{\sqrt{\cos 2t}}, y = \frac{\cos^3 t}{\sqrt{\cos 2t}}$

(vi) $x = a \cos \theta, y = b \cos \theta$

(vii) $x = \cos \theta - \cos 2\theta, y = \sin \theta - \sin 2\theta$

(viii) $x = a(\theta - \sin \theta), y = a(1 + \cos \theta)$

(ix) $x = a \left(\cos t + \log \tan \frac{t}{2} \right), y = a \sin t$

(x) $x = a \sec \theta, y = b \tan \theta$

(xi) $x = a(\cos \theta + \theta \sin \theta), y = a(\sin \theta - \theta \cos \theta)$

Q.2 If $x = \sqrt{a \sin^{-1} t}, y = \sqrt{a \cos^{-1} t}$, show that $\frac{dy}{dx} = -\frac{y}{x}$.

Q.3 Find $\frac{d^2y}{dx^2}$, if $y = x^3 + \tan x$.

Q.4 (i) If $y = 3e^{2x} + 2e^{3x}$, prove that $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = 0$.

(ii) If $y = \sin^{-1}x$, show that $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 0$.

(iii) If $y = 5 \cos x - 3 \sin x$, prove that $\frac{d^2y}{dx^2} + y = 0$.

(iv) If $y = 3 \cos (\log x) + 4 \sin (\log x)$, show that $x^2 y_2 + x y_1 + y = 0$

(v) If $y = Ae^{mx} + Be^{nx}$, show that $\frac{d^2y}{dx^2} - (m+n) \frac{dy}{dx} + mny = 0$

(vi) If $e^y (x+1) = 1$. Show that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx} \right)^2$

(vii) If $y = (\tan^{-1}x)^2$, show that $(x^2 + 1)^2 y_2 + 2x(x^2 + 1) y_1 = 2$.

Q.6 For a positive constant a find $\frac{dy}{dx}$, where $y = a^{t + \frac{1}{t}}$ and $x = \left(t + \frac{1}{t} \right)^a$.

Q.7 Differentiate $\sin^2 x$ w.r.t. $e^{\cos x}$.

Q.8 Find $\frac{dy}{dx}$, if $y = 12(1 - \cos t), x = 10(t - \sin t), -\frac{\pi}{2} < t < \frac{\pi}{2}$.

Q.9 Find $\frac{dy}{dx}$, if $y = \sin^{-1}x + \sin^{-1}\sqrt{1-x^2}$, $-1 \leq x \leq 1$

Q.10 If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, for $-1 < x < 1$, prove that $\frac{dy}{dx} = -\frac{1}{(1+x)^2}$.

Q.11 If $\cos y = x \cos(a+y)$, with $\cos a \neq \pm 1$, prove that $\frac{dy}{dx} = \frac{\cos^2(a+y)}{\sin a}$.

Q.12 If $x = a(\cos t + t \sin t)$ and $y = a(\sin t - t \cos t)$, find $\frac{d^2y}{dx^2}$.

Q.13 If $y = e^{a \cos^{-1}x}$, $-1 \leq x \leq 1$, show that $(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - a^2y = 0$.

Q.14 If $y = \begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix}$, find $\frac{dy}{dx}$.

Q.15 If $f(x) = \begin{vmatrix} x^n & n! & 2 \\ \cos x & \cos \frac{n\pi}{2} & 4 \\ \sin x & \sin \frac{n\pi}{2} & 8 \end{vmatrix}$, then find the value of $\frac{d^n}{dx^n} [f(x)]_{x=0}$

Answer key

Q.1 (i) $-\cot \theta$ (ii) $\frac{1}{t}$ (iii) $-\left(\frac{y}{x}\right)^{1/3}$ (iv) $t^2(v) - \cot 3t$

(vi) $\frac{b}{a}$ (vii) $\frac{(\cos \theta - 2 \cos 2\theta)}{(2 \sin 2\theta - \sin \theta)}$ (viii) $-\cot\left(\frac{\theta}{2}\right)$ (ix) $\tan t$

(x) $\frac{b}{a} \operatorname{cosec} \theta$ (xi) $\tan \theta$

Q.3 $6x + 2 \sec^2 x \tan x$ **Q.6** $\frac{a^{\frac{1}{t} + \frac{1}{t} \log a}}{a\left(t + \frac{1}{t}\right)^{a-1}}$ **Q.7** $-\frac{2 \cos x}{e^{\cos x}}$ **Q.8** $\frac{6}{5} \cot\left(\frac{t}{2}\right)$

Q.9 0 **Q.12** $\frac{\sec^3 t}{at}$, $0 < t < \frac{\pi}{2}$ **Q.14** 1 **Q.15** 0

DEDUCTION OF NEW IDENTITIES BY DIFFERENTIATING A GIVEN IDENTITY :

Illustration :

If $\cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \cdots \cos \frac{x}{2^n} = \frac{\sin x}{2^n \sin(x/2^n)}$, then prove that

$$\sum_{r=1}^n \frac{1}{2^r} \tan \frac{x}{2^r} = \frac{1}{2^n} \cot \frac{x}{2^n} - \cot x$$

Sol. $\cos \frac{x}{2} \cdot \cos \frac{x}{2^2} \cdot \cos \frac{x}{2^3} \cdots \cos \frac{x}{2^n} = \frac{\sin x}{2^n \sin(x/2^n)} \quad \dots\dots(1)$

Taking logarithm of both sides

$$\log \left(\cos \frac{x}{2} \right) + \log \left(\cos \frac{x}{2^2} \right) + \log \left(\cos \frac{x}{2^3} \right) + \dots\dots + \log \left(\cos \frac{x}{2^n} \right) = \log (\sin x) - \log \left(2^n \sin \left(\frac{x}{2^n} \right) \right)$$

Differentiating w.r.t. to x

$$-\left(\frac{1}{2} \frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} + \frac{1}{2^2} \frac{\sin \left(\frac{x}{2^2} \right)}{\cos \left(\frac{x}{2^2} \right)} + \dots\dots + \frac{1}{2^n} \frac{\sin \left(\frac{x}{2^n} \right)}{\cos \left(\frac{x}{2^n} \right)} \right) = \cot x - \frac{1}{2^n \sin \left(\frac{x}{2^n} \right)} \cdot \cos \left(\frac{x}{2^n} \right)$$

so $\sum_{r=1}^n \frac{1}{2^r} \tan \left(\frac{x}{2^r} \right) = \frac{1}{2^n} \cot \left(\frac{x}{2^n} \right) - \cot x.$

Illustration :

If $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots\dots + C_nx^n$, then prove that

$$C_1 + 2C_2 + 3C_3 + \dots\dots + nC_n = n2^{n-1}$$

and $C_0 + 2C_1 + 3C_2 + \dots\dots + (n+1)C_n = (n+2)2^{n-1}$

Sol. Given $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots\dots + C_nx^n \quad \dots\dots\dots(1)$

(i) Differentiating equation (1) w.r. to x

$$n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \dots\dots + nC_nx^{n-1}.$$

put $x = 1$

$$n2^{n-1} = C_1 + 2C_2 + 3C_3 + \dots\dots + nC_n$$

(ii) Now multiply x on both sides of equation (1)

$$x(1+x)^n = C_0x + C_1x^2 + C_2x^3 + \dots\dots + C_nx^{n+1}$$

Differentiating equation (1) w.r.t to x ,

$$(1+x)^n + nx(1+x)^{n-1} = C_0 + 2C_1x + 3C_2x^2 + \dots\dots + (n+1)C_nx^n$$

put $x = 1$

$$2^n + n2^{n-1} = C_0 + 2C_1 + 3C_2 + \dots\dots + (n+1)C_n$$

$$2^{n-1}(n+2) = C_0 + 2C_1 + 3C_2 + \dots\dots + (n+1)C_n$$

DERIVATIVE OF FUNCTIONS EXPRESSED IN THE DETERMINANT FORM :

Let $F(x) = \begin{vmatrix} f & g & h \\ u & v & w \\ l & m & n \end{vmatrix}$ where all functions are differentiable then

$$D'(x) = \begin{vmatrix} f' & g' & h' \\ u & v & w \\ l & m & n \end{vmatrix} + \begin{vmatrix} f & g & h \\ u' & v' & w' \\ l & m & n \end{vmatrix} + \begin{vmatrix} f & g & h \\ u & v & w \\ l' & m' & n' \end{vmatrix}$$

This result may be proved by first principle and the same operation can also be done column wise.

Illustration :

If f, g & h are differentiable functions of x & $D = \begin{vmatrix} f & g & h \\ (xf)' & (xg)' & (xh)' \\ (x^2f)'' & (x^2g)'' & (x^2h)'' \end{vmatrix}$

prove that $D' = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ (x^3f'')' & (x^3g'')' & (x^3h'')' \end{vmatrix}$

Sol. $D = \begin{vmatrix} f & g & h \\ (xf)' & (xg)' & (xh)' \\ (x^2f)'' & (x^2g)'' & (x^2h)'' \end{vmatrix}$

$$D = \begin{vmatrix} f & g & h \\ f + xf' & g + xg' & h + xh' \\ 2f + 4xf' + x^2f'' & 2g + 4xg' + x^2g'' & 2h + 4xh' + x^2h'' \end{vmatrix}$$

After doing row and column operation. This determinant simplifies to

$$D = \begin{vmatrix} f & g & h \\ xf' & xg' & xh' \\ x^2f'' & x^2g'' & x^2h'' \end{vmatrix} = x^3 \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix}$$

$$D = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ x^3f'' & x^3g'' & x^3h'' \end{vmatrix}$$

$$D' = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ (x^3f'')' & (x^3g'')' & (x^3h'')' \end{vmatrix}$$

Illustration :

If $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$ then find $f'(x)$.

Sol. $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$

Expanding the determinant

$$\begin{aligned} f(x) &= \cos(x+x^2) \cos(x-x^2) \sin 2x^2 - \sin(x+x^2) \sin(x-x^2) \sin 2x \\ &\quad + \cos(x+x^2) \cos(x-x^2) \sin 2x \\ &= \sin 2x^2 \cos 2x + \sin 2x \cos(2x^2) = \sin(2x^2 + 2x) \\ f'(x) &= \cos(2x^2 + 2x) 2(2x + 1) \end{aligned}$$

Illustration :

If α be a repeated root of a quadratic equation $f(x) = 0$ & $A(x)$, $B(x)$, $C(x)$ be the polynomials

of degree 3, 4 & 5 respectively, then show that $\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ is divisible by $f(x)$, where dash denotes the derivative.

Sol. $g(x) = \begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ and $g'(x) = \begin{vmatrix} A'(x) & B'(x) & C'(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$

so $g(\alpha) = 0$ and $g'(\alpha) = 0$

so α is a repeated root of $g(x)$.

so $g(x) = (x - \alpha)^2 h(x) = f(x) h(x)$

so $g(x)$ is divisible by $f(x)$.

Note : If $(x - r)$ is a factor of the polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_0$ repeated m times where $1 \leq m \leq n$ then r is a root of the equation $f'(x) = 0$ repeated $(m - 1)$ times.

General things to remember :

Illustration :

Prove that the derivative of an even differentiable function is an odd function and the derivative of an odd differentiable function is an even function.

Sol. Let $f(x)$ be even function

$$f(-x) = f(x)$$

$$f'(-x) (-1) = f'(x)$$

$$\Rightarrow f'(-x) = -f'(x) \Rightarrow f'(x) \text{ is an odd function.}$$

Let $f(x)$ be odd function

$$f(-x) = -f(x)$$

$$f'(-x) (-1) = -f'(x)$$

$$\Rightarrow f'(-x) = f'(x) \Rightarrow f'(x) \text{ is an even function.}$$

3.0 L'HOSPITAL'S RULE ($0^0 / \infty^0$):

e.g. $f(x) = x^x$ or $\left(-\frac{1}{x}\right)^{\sin x}$

If $f(x)$ and $g(x)$ are two function such that

- (i) $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$
- (ii) f and g are derivable / continuous at $x = a$
i.e. $\lim_{x \rightarrow a} f(x) = f(a) = 0$; $\lim_{x \rightarrow a} g(x) = g(a) = 0$
- (iii) $f'(x)$ and $g'(x)$ are continuous at $x = a$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)}$$

Illustration :

Evaluate $\lim_{x \rightarrow 0} \frac{x \cos x - \ln(1+x)}{x^2}$.

Sol. $\lim_{x \rightarrow 0} \frac{x \cos x - \ln(1+x)}{x^2} \quad \left(\frac{0}{0}\right) \text{ form}$

$$= \lim_{x \rightarrow 0} \frac{(\cos x - x \sin x) - \frac{1}{1+x}}{2x} \quad \left(\frac{0}{0}\right) \text{ form}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin x - (\sin x + x \cos x) + \frac{1}{(1+x)^2}}{2} = \frac{-0 - (0+0) + 1}{2} = \frac{1}{2}.$$

Illustration :

Evaluate find a and b if $\lim_{x \rightarrow 0} \frac{x(1+a \cos x) - b \sin x}{x^3} = 1$

Sol. Let $L = \lim_{x \rightarrow 0} \frac{x(1+a \cos x) - b \sin x}{x^3} \quad \left(\frac{0}{0}\right) \text{ form}$

$$L = \lim_{x \rightarrow 0} \frac{(1+a \cos x) - ax \sin x - b \cos x}{3x^2} \quad \left(\frac{0}{0}\right) \text{ form}$$

$$1 + a - 0 - b = 0 \Rightarrow b - a = 1 \quad \dots\dots\dots(1)$$

$$L = \lim_{x \rightarrow 0} \frac{-a \sin x - a(x \cos x + \sin x) + b \sin x}{6x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{6} \left[-a \left(\frac{\sin x}{x} \right) - a \cos x - \frac{a \sin x}{x} + \frac{b \sin x}{x} \right] = \frac{1}{6} (-a - a - a + b) = 1$$

$$\Rightarrow -3a + b = 6 \quad \dots\dots\dots(2)$$

From equation (1) and (2)

$$a = \frac{-5}{2}, b = \frac{-3}{2}.$$

Illustration :

$$\lim_{x \rightarrow 0} \frac{e^x \sin x - x - x^2}{x^2 + x - \ln(1-x)}$$

Sol. $L = \lim_{x \rightarrow 0} \frac{e^x \sin x - x - x^2}{x^2 + x - \ln(1-x)} \quad \left(\frac{0}{0}\right) \text{ form}$

$$= \lim_{x \rightarrow 0} \frac{e^x(\cos x + \sin x) - 1 - 2x}{2x + 1 + \frac{1}{(1-x)}} \quad \left(\frac{0}{0}\right) \text{ form}$$

$$= \frac{1-1-0}{0+1+1} = 0.$$

Illustration :

Evaluate $\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \ln(1-x)}{x \tan^2 x}$

(A) $-\frac{1}{2}$

(B) $-\frac{1}{3}$

(C) $\frac{1}{6}$

(D) DNE

Sol. $\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \ln(1-x)}{x \tan^2 x}$

$$= \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \ln(1-x)}{x^3 \left(\frac{\tan^2 x}{x^2} \right)}$$

$$= \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \ln(1-x)}{x^3} \quad \left(\frac{0}{0}\right) \text{ form}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x + \sin x - \frac{1}{1-x}}{3x^2} \quad \left(\frac{0}{0}\right) \text{ form}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin x + \cos x - \frac{1}{(1-x)^2}}{6x} \quad \left(\frac{0}{0}\right) \text{ form}$$

$$= \lim_{x \rightarrow 0} \frac{-\cos x + \sin x - \frac{2}{(1-x)^3}}{6} = \frac{-1+0-2}{6} = \frac{-1}{2}.$$

Illustration :

Evaluate $\lim_{x \rightarrow 0} \frac{\log_{\sec \frac{x}{2}}(\cos x)}{\log_{\sec x}(\cos(\frac{x}{2}))}$
(A) 1 (B) 16 (C) 4 (D) 2

Sol.
$$\lim_{x \rightarrow 0} \frac{\log_{\sec \frac{x}{2}} \cos x}{\log_{\sec x} \cos \left(\frac{x}{2}\right)} = \lim_{x \rightarrow 0} \frac{\left(\frac{\ln \cos x}{\ln \sec \frac{x}{2}}\right)}{\left(\frac{\ln \cos \frac{x}{2}}{\ln \sec x}\right)} = \lim_{x \rightarrow 0} \left(\frac{\ln \cos x}{\ln \cos \frac{x}{2}}\right)^2$$

$$= \left(\lim_{x \rightarrow 0} \frac{\ln \cos x}{\ln \cos \left(\frac{x}{2}\right)}\right)^2 \left(\frac{0}{0}\right) \text{ form} = \left(\lim_{x \rightarrow 0} \frac{-\tan x}{\frac{-1}{2} \tan \frac{x}{2}}\right)^2 = \left(\lim_{x \rightarrow 0} 4 \frac{\frac{\tan x}{x}}{\frac{\tan \frac{x}{2}}{\frac{x}{2}}}\right)^2 = 4^2 = 16.$$

Illustration :

Evaluate $\lim_{x \rightarrow 1} \frac{x^x - x}{x - 1 - \ln x}$

Sol.
$$\lim_{x \rightarrow 1} \frac{x^x - 1}{x - 1 - \ln x} = \lim_{x \rightarrow 1} \frac{e^{x \ln x} - x}{x - 1 - \ln x} \left(\frac{0}{0}\right) \text{ form}$$

$$\lim_{x \rightarrow 1} \frac{e^{x \ln x} (1 + \ln x) - 1}{\left(1 - \frac{1}{x}\right)} \left(\frac{0}{0}\right) \text{ form}$$

$$= \lim_{x \rightarrow 1} \frac{e^{x \ln x} (1 + \ln x)^2 + e^{x \ln x} \cdot \frac{1}{x} - 0}{\left(\frac{1}{x^2}\right)} \left(\frac{0}{0}\right) \text{ form}$$

$$= 1 + 1 - 0 = 2.$$

Illustration :

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x \cdot \cos 2x \cdot \cos 3x}{x^2}$

Sol.
$$\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x \cos 3x}{x^2} = \lim_{x \rightarrow 0} \frac{1 - e^{(\ln \cos x + \ln \cos 2x + \ln \cos 3x)}}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{-e^{(\ln \cos x + \ln \cos 2x + \ln \cos 3x)} \cdot (-\tan x - 2 \tan 2x - 3 \tan 3x)}{2x} = \frac{e^0}{2} (1 + 4 + 9) = 7. \text{ Ans.}$$

Illustration :

Evaluate $\lim_{x \rightarrow 0} \frac{(\cos ax)^{1/m} - (\cos bx)^{1/n}}{x^2}$

Sol. $\lim_{x \rightarrow 0} \frac{(\cos ax)^{1/m} - (\cos bx)^{1/n}}{x^2} \left(\frac{0}{0} \right) \text{ form}$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{m} (\cos ax)^{\frac{1}{m}-1} (-a \sin ax) - \frac{1}{n} (\cos bx)^{\frac{1}{n}-1} (-b \sin bx)}{2x}$$

$$= \frac{1}{2} \left[\frac{-a}{m} \lim_{x \rightarrow 0} (\cos ax)^{\frac{1}{m}-1} \left(\frac{\sin ax}{x} \right) + \frac{b}{n} \lim_{x \rightarrow 0} (\cos bx)^{\frac{1}{n}-1} \frac{\sin bx}{x} \right]$$

$$= \frac{1}{2} \left[\frac{-a}{m} \cdot a + \frac{b}{n} \cdot b \right] = \frac{1}{2} \left(\frac{b^2}{n} - \frac{a^2}{m} \right) = \left(\frac{mb^2 - na^2}{2mn} \right).$$

Note : If $\lim_{x \rightarrow a} f(x) \rightarrow \infty$ and $\lim_{x \rightarrow a} g(x) \rightarrow \infty$ then also $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$

Illustration :

Evaluate the following limits

(i) $\lim_{x \rightarrow 0^+} (\operatorname{cosec} x)^{\frac{1}{\ln x}}$ (ii) $\lim_{x \rightarrow \pi/2} (\sec x)^{\cot x}$ (iii) $\lim_{x \rightarrow 0^+} (\cot x)^{\frac{1}{\ln x}}$

(iv) $\lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{\tan x}$ (v) $\lim_{x \rightarrow 0} x^x$ (vi) $\lim_{x \rightarrow 1} (1-x^2)^{\frac{1}{\ln(1-x)}}$

(vii) $\lim_{x \rightarrow 4} \frac{\sqrt{2x+1} + \sqrt{x-3} - 4}{\sqrt{(3x+4)} + \sqrt{5x+5} - 9}$

Sol.

(i) $L = \lim_{x \rightarrow 0^+} (\operatorname{cosec} x)^{\frac{1}{\ln x}}$

$$\ln L = \lim_{x \rightarrow 0^+} \frac{\ln \operatorname{cosec} x}{\ln x} \left(\frac{\infty}{\infty} \right) \text{ form}$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{\operatorname{cosec} x} (-\operatorname{cosec} x \cdot \cot x)}{\left(\frac{1}{x} \right)} = \lim_{x \rightarrow 0^+} - \left(\frac{x}{\tan x} \right) = -1.$$

$$= L = e^{-1}$$

$$(ii) \quad L = \lim_{x \rightarrow \frac{\pi}{2}} (\sec x)^{\cot x}$$

$$\ln L = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\ln \sec x}{\tan x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{1}{\sec x} \cdot (\sec x \cdot \tan x)}{\sec^2 x} = \lim_{x \rightarrow \frac{\pi}{2}} \sin x \cdot \cos^2 x = 0$$

$$L = e^0 = 1.$$

$$(iii) \quad L = \lim_{x \rightarrow 0^+} (\cot x)^{\frac{1}{\tan x}} \Rightarrow \ln L = \lim_{x \rightarrow 0^+} \frac{\ln(\cot x)}{\ln x}$$

$$\ln L = \lim_{x \rightarrow 0^+} \frac{\frac{1}{\cot x} (-\operatorname{cosec}^2 x)}{\left(\frac{1}{x}\right)} = \lim_{x \rightarrow 0^+} \frac{-x}{\left(\frac{\cos x}{\sin x} \cdot \sin^2 x\right)} = \lim_{x \rightarrow 0^+} \frac{-2x}{\sin 2x} = -1$$

$$\Rightarrow L = \frac{1}{e}.$$

$$(iv) \quad L = \lim_{x \rightarrow 0} \left(\frac{1}{x}\right)^{\tan x}$$

$$\ln L = \lim_{x \rightarrow 0} \tan x (-\ln x) = \lim_{x \rightarrow 0} \frac{-\ln x}{\cot x} \left(\frac{\infty}{\infty}\right) \text{ form}$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{x}}{-\operatorname{cosec}^2 x} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x}\right) \cdot \sin x = 0$$

$$L = 1.$$

$$(v) \quad L = \lim_{x \rightarrow 0} x^x$$

$$\ln L = \lim_{x \rightarrow 0} x \ln x = \lim_{x \rightarrow 0} \frac{\ln x}{(1/x)} \left(\frac{\infty}{\infty}\right) \text{ form}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{x}}{\left(\frac{-1}{x^2}\right)} = \lim_{x \rightarrow 0} (-x) = 0$$

$$L = 1.$$

$$(vi) \quad L = \lim_{x \rightarrow 1} (1-x^2)^{\frac{1}{1-x}}$$

$$\ln L = \lim_{x \rightarrow 1} \frac{\ln(1-x^2)}{\ln(1-x)} \quad \left(\frac{\infty}{\infty}\right) \text{ form}$$

$$= \lim_{x \rightarrow 1} \frac{\left(\frac{-2x}{1-x^2}\right)}{\left(\frac{-1}{1-x}\right)} = \lim_{x \rightarrow 1} \frac{2x}{1+x} = 1.$$

$$L = e$$

$$\begin{aligned}
 \text{(vii)} \quad & \lim_{x \rightarrow 4} \frac{\sqrt{2x+1} + \sqrt{x-3} - 4}{\sqrt{3x+4} + \sqrt{5x+5} - 9} \quad \left(\frac{0}{0} \right) \text{ form} \\
 &= \lim_{x \rightarrow 4} \frac{\frac{2}{2\sqrt{2x+1}} + \frac{1}{2\sqrt{x-3}} - 0}{\frac{3}{2\sqrt{3x+4}} + \frac{5}{2\sqrt{5x+5}} - 0} \left(\frac{0}{0} \right) \text{ form} \\
 &= \frac{\frac{2}{3} + \frac{1}{4}}{\frac{3}{4} + 1} = \frac{\frac{5}{4}}{\frac{7}{4}} = \left(\frac{20}{21} \right).
 \end{aligned}$$

Practice Problem

Q.1 Evaluate the following limits using L'hospital's Rule

- (i) $\lim_{x \rightarrow \frac{\pi}{2}} \tan x \log \sin x$ (ii) $\lim_{x \rightarrow 0} \frac{\log \cos x}{x}$
- (iii) $\lim_{x \rightarrow 0} \frac{2^x - 1}{(1+x)^{1/2} - 1}$ (iv) $\lim_{x \rightarrow \frac{\pi}{4}} (2 - \tan x)^{\frac{1}{\ln(\tan x)}}$

Q.2 If $\lim_{x \rightarrow a} \frac{a^x - x^a}{x^x - a^a} = -1$ and $a > 0$, then find the value of a .

Q.3 If $f(x) = \frac{\ln(x^2 + e^x)}{\ln(x^4 + e^{2x})}$. Then $\lim_{x \rightarrow \infty} f(x)$ is equal to

- (A) 1 (B) 1/2 (C) 2 (D) None of these

Q.4 Let $f(x+y) = f(x) \cdot f(y)$ for all x and y . Suppose $f(5) = 2$ and $f'(0) = 3$, find $f'(5)$.

Q.5 Let $f(xy) = f(x) f(y) \forall x, y \in \mathbb{R}$ and f is differentiable at $x = 1$ such that $f'(1) = 1$ also $f(1) \neq 0$, $f(2) = 3$, then find $f'(2)$.

Q.6 If $f\left(\frac{x+2y}{3}\right) = \frac{f(x)+2f(y)}{3} \forall x, y \in \mathbb{R}$ and $f'(0) = 1$, $f(0) = 2$, then find $f(x)$.

Answer key

- | | | | | | |
|------------|---------|--------|---------------|---------------|--------------|
| Q.1 | (i) 0 | (ii) 0 | (iii) $\ln 4$ | (iv) e^{-1} | |
| Q.2 | $a = 1$ | | Q.3 | B | Q.4 6 |
| Q.5 | 6 | | Q.6 | $x+2$ | |